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The CRAFT(-ED) Framework for Effective AI-Augmented Writing: Navigating the Inside-Outside Paradox with Multiplexity

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ABSTRACT In the era of generative AI (GenAI), writers face a genuine dilemma. Relying on AI too heavily yields fluent text that can outrun the writer's own understanding, producing performance without formation; rejecting it outright forfeits real gains in clarity, efficiency, and reach. Neither pole suffices, so effective AI-augmented writing demands a dynamic balance between them. We argue that *polarity navigation*, the practice of managing an enduring tension rather than solving it once, offers a productive lens for striking this balance. At the heart of the dilemma lies a craft-versus-performance paradox that we frame as an *inside-outside paradox*: AI can manufacture polished artifacts without genuine comprehension (outside without inside), while deep understanding can fail to reach readers (inside without outside). To navigate it, we propose CRAFT(-ED), a *multiplex* framework that deliberately holds two complementary poles together rather than choosing between them. Multiplexity here is more than both-and thinking: it reads the problem along three dimensions, horizontal (holding the poles), vertical (ranking formation above fluency), and dynamic (tracking their balance over time). Inside-focused *CRAFT* (Consciousness, Relational, Agentic, Formative, Truthful) cultivates the writer's formation; outside-focused *CRAFTED* (Coherence, Reader Focus, Accuracy, Flow, Top-Down Thinking, Espirited Authenticity, Directed Design) governs the quality of the writing; their integration is the CRAFT(-ED) framework. Methodologically, the framework is an integrative synthesis of polarity management theory, capability-development philosophy, and writing pedagogy, offered as a Perspective to be empirically validated in follow-up work.

INDEX TERMS Generative AI, AI-Augmented Writing, Polarity Management, Human-AI Collaboration.

I. Introduction: When Anyone Can Generate, Learn the CRAFT

> "*The central virtue of a liberal education is that it teaches you how to write, and writing makes you think. Whatever you do in life, the ability to write clearly, cleanly, and reasonably quickly will prove to be an invaluable skill*."—Fareed Zakaria, *In Defense of A Liberal Education* [1]

Writing has indeed become invaluable in the generative AI (GenAI) era, but for a paradoxical reason. When anyone can generate grammatically correct text at the click of a button, the central challenge shifts from production to discernment: text generation becomes commodified, while authentic writing, the kind that reflects genuine understanding and disciplined thought, remains irreplaceable. This concern was crystallized in 2025 when the Cambridge Dictionary selected "slop" as its Word of the Year, defining it as "content on the internet that is of very low quality, especially when it is created by AI" [2]. The diagnosis was independently reinforced by Merriam-Webster, which likewise identified "slop" as emblematic of the growing prevalence of unreflective AI-generated content and its cultural and epistemic consequences [3].

The proliferation of AI-generated "slop" captures our defining paradox: we can produce polished surfaces without developing genuine capability, or possess deep understanding yet fail to communicate it effectively. Excellence in AI-augmented writing requires navigating this inside-outside tension, not resolving it. The framework we develop for this purpose, CRAFT(-ED), joins two halves: *CRAFT*, the inside work of writer formation, and *CRAFTED*, the outside discipline of producing effective text. Their integration, written CRAFT(-ED), is what we mean by a *multiplex* framework: one that holds both halves together rather than trading one for the other.

A. Inside-Outside Paradox: Education's Defining Challenge

The advent of GenAI presents education with what we term the *inside-outside paradox*: the growing disconnect between internal understanding and external artifacts in academic writing. Each pole shown in Figure 1 carries distinct risks when pursued in isolation, generating two complementary failure modes that waste human potential while missing AI's transformative capabilities.

1) Outside without Inside

AI-generated text produces impressive artifacts (polished, grammatically correct, structurally sound documents) without genuine understanding. This is AI slop: counterfeit scholarship that appears credible but lacks authentic insight, distinctive

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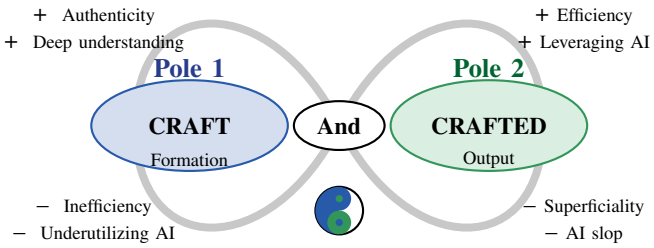


Fig. 1: **The CRAFT-CRAFTED Polarity.** *Neither formation (CRAFT) nor output focus (CRAFTED) suffices alone. Excellence requires both-and navigation: authentic development with effective communication, leveraging AI technology effectively without losing authenticity. This is managed through the CRAFT(-ED) framework.*

voice, or deep comprehension. The writer accumulates output without mastery, performance without capability.

2) Inside without Outside

Deep expertise fails to communicate effectively, remaining trapped behind poor organization, inaccessible language, or underuse of available tools. This represents wasted potential: genuine understanding that never reaches, influences, or benefits its intended audience.

This paradox matters profoundly for education because writing is not merely a communication skill but a formative practice. When AI short-circuits the developmental struggle that transforms writers, it lets students produce without becoming, generating outputs without developing the underlying capabilities (a point we develop, with Sennett, in Section III).

Yet completely rejecting AI perpetuates persistent human weaknesses: unclear thinking, poor organization, inaccessible language, and inefficient processes that limit rather than enable authentic scholarship. The romantic notion of "pure" human work ignores that unassisted writing often suffers from cognitive biases, limited perspectives, and unnecessary barriers to clarity [4].

A millennium-old episode sharpens the point. When robbers stole the manuscripts of the philosopher Al-Ghazali, one thief mocked him: "True knowledge cannot be stolen; it must live within you" [5], [6]. Al-Ghazali had mistaken external artifacts for internal understanding, precisely the inside-outside distinction at issue here: the manuscripts then (AI-generated documents now) are external artifacts, while memorized knowledge then (genuine comprehension now) is internal capability. His response was instructive. Rather than rejecting writing technology or relying on it uncritically, he wrote both *to inform* (external documentation) and *to remember* (internalized understanding), so that the tool extended his capability without replacing it [5]. Today's writers face the same task: producing impressive AI-assisted texts without letting their own understanding go shallow.

B. The Both-And CRAFT(-ED) Multiplex Framework

We propose that this inside-outside paradox cannot be solved but must be navigated through *multiplex thinking*: maintaining both dimensions simultaneously rather than choosing between them. Multiplexity is a construct we have developed in earlier work on integrative, pluralistic reasoning [7], [8], which we adapt here to AI-augmented writing. Building on polarity management theory [9], [10], which distinguishes between problems to solve and tensions to navigate, we introduce the CRAFT(-ED) framework as a multiplex model that integrates two complementary poles (see Figure 1):

CRAFT (*inside-focused*): Five dimensions of writer formation and capability development, namely Consciousness, Relational, Agentic, Formative, and Truthful. This pole addresses the question: Is the writer developing genuine understanding and authentic voice?

CRAFTED (outside-focused): Seven pillars of effective communication and reader-focused outputs, namely Coherence, Reader Focus, Accuracy, Flow, Top-Down Thinking, Espirited Authenticity, and Directed Design. This pole addresses the question: Is the work clear, accessible, and valuable to readers?

Excellence emerges from simultaneously maintaining both dimensions. *Inside-focused CRAFT* dimensions (Consciousness, Relational, Agentic, Formative, Truthful) are directed at authentic development. *Outside-focused CRAFTED* pillars (Coherence, Reader Focus, Accuracy, Flow, Top-Down, Espirited, Directed) are directed at effective communication. *The yin-yang symbol underneath the center box represents their necessary interdependence*.

The multiplex CRAFT(-ED) model emerges from integrating these poles through both-and thinking. Neither optimizing solely for polished outputs (risking AI slop and shallow understanding) nor focusing exclusively on capability development (risking communication failure and hidden expertise) suffices alone. As Postman observes, "every technology is both a burden and a blessing; not either-or, but this-and-that" [11]. Authentic AI-augmented scholarship requires a dynamic equilibrium between inside and outside, avoiding three distinct forms of waste: the commodified homogeneity of AI over-reliance, the persistent inefficiency of AI rejection, and the missed transformative potential of both extremes.

C. Beyond Slop: The Path Forward

Recent evidence makes the urgency of this challenge concrete, and it pulls in two directions. One body of work documents the cognitive costs of over-reliance. Messeri and Crockett [12] warn that AI tools can foster "illusions of understanding" in science, a culture that produces more while comprehending less, a concern they later sharpen for large language models [13]. In a similar spirit, Zohar, Bloom, and Inzlicht [14] argue that the friction AI removes is often developmentally necessary, since the struggles that build competence and make achievements feel earned are precisely what frictionless tools dissolve. The pattern is already visible in practice: organizations report a flood of low-quality "workslop" that depresses rather than lifts productivity [15], students accumulate "cognitive debt" when they lean on AI assistants [16], and reviews of automation find measurable skill atrophy [17].

A second body of work points the other way. Meta-analyses by Liu et al. [18] and Deng et al. [19] find that appropriately scaffolded AI use can improve learning outcomes. The two findings appear to conflict, but the conflict dissolves once AI

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integration is read not as a binary choice but as a persistent polarity: the same tool helps or harms depending on how its use is balanced, so the task is one of continuous navigation rather than one-time judgment.

Yet institutional responses have largely favored technological fixes, plagiarism detection, automated quality metrics, and productivity dashboards, over the harder work of pedagogical transformation. This *techno-solutionism*, the assumption that complex educational and ethical problems yield to technical fixes alone [20], addresses what is easiest to monitor while skirting the questions that matter: are students transforming AI-generated material into authentic intellectual work or merely polishing surfaces, and are they cultivating genuine cognitive capability or outsourcing it? Answering these requires revised assessment philosophies, new scaffolding strategies, and a rethinking of what it means to learn with AI, work that is harder than deploying detection software but necessary if we are to realize AI's genuine educational promise.

The CRAFT(-ED) framework offers a path forward by making the invisible visible. Rather than external detection of AI use, we offer internal discovery tools for writers to assess both their artifacts (outside) and their capabilities (inside). Rather than binary choices between human effort and AI assistance, we provide polarity maps for navigating dynamic equilibrium. Rather than treating AI integration as a problem to solve, we recognize it as a persistent tension requiring ongoing wisdom, what polarity management theory identifies as a "divergent problem" [9], [21].

D. Contribution and Structure

This paper makes three primary contributions to the challenge of authentic AI-augmented writing:

Theoretical Integration. We provide a comprehensive synthesis of (a) polarity management theory, (b) capability development philosophy, and (c) practical writing pedagogy, specifically addressing the inside-outside paradox in GenAI contexts; to our knowledge, this integration is among the first to bring these traditions together for AI-augmented writing. This integration explains why avoiding slop requires navigating persistent tensions rather than implementing fixed solutions.

Multiplex Framework Architecture. The CRAFT(-ED) model operates as a unified system linking inner formation (CRAFT dimensions) to outer expression (CRAFTED pillars). Its distinctiveness lies less in any single borrowed theory than in reading AI-augmented writing through the three dimensions of multiplexity together, horizontal both-and reasoning, vertical value-ordering that gives priority to formation over fluency, and dynamic feedback over time, a combination that, to our knowledge, has not previously been brought to bear on this problem. By grounding each pillar in its underlying polarities, the framework is intended to help writers harness AI without being absorbed into generic, commodified patterns. Section III establishes the theoretical foundations, Section IV details the five CRAFT dimensions, and Section V presents all seven CRAFTED pillars as an integrated system.

Practical Navigation Tools. We provide self-assessment instruments, polarity maps, concrete prompts, and evidence-based guidance that move beyond detection toward discovery. These tools are intended to help writers evaluate both their outputs and their developmental journey and to maintain dynamic equilibrium between inside authenticity and outside effectiveness; as exploratory instruments grounded in prior theory rather than validated measures, their benefits are put forward as plausible and evidence-motivated rather than demonstrated.

The remainder of this paper is organized as follows. Section II describes our research approach and methodology. Section III establishes the theoretical foundations in polarity management and multiplexity. Section IV introduces the five CRAFT dimensions for inside-focused capability development, and Section V presents the seven CRAFTED pillars for outside-focused communication, each enhanced with polarity maps and self-assessment tools. Section VI examines ethical considerations through a polarity lens. Section VII then positions CRAFT(-ED) within existing scholarship, and Section VIII synthesizes implications for research, practice, and design. We conclude in Section IX with future directions for research and practice.

II. Research Approach and Methodology

This paper is a *Perspective* paper that presents a normative conceptual framework grounded in evidence-based synthesis. We describe our research approach, its methodological rationale, and its scope and limitations below.

A. Nature of the Contribution

The CRAFT(-ED) framework is a work of *conceptual analysis and theoretical synthesis*: it integrates established traditions, namely polarity management theory [9], [10], capability development philosophy [22]–[24], paradox theory [25], [26], and writing pedagogy [27]–[29], in dialogue with the empirical literature on AI-augmented writing [30]–[32], into a novel framework for navigating the inside-outside paradox. Such work follows a well-established tradition in the social sciences and education research, where conceptual frameworks must be articulated before they can be operationalized and empirically tested [33].

Theoretical synthesis of this kind is itself a recognized contribution, distinct from but complementary to empirical research: by organizing existing knowledge around a unifying problem, naming the tensions prior work has left unaddressed, and proposing shared constructs and vocabulary, a conceptual framework builds the infrastructure that focused empirical work later requires [33]. The approach has clear precedent: Messeri and Crockett [12] advanced a taxonomy of AI illusions in *Nature* through conceptual analysis without new data, and Zohar, Bloom, and Inzlicht [14] argued against frictionless AI in *Communications Psychology* through theoretical synthesis. We are equally explicit about what CRAFT(-ED) does not claim: its self-assessment instruments and polarity maps are exploratory tools that reflect theoretical reasoning, not measured outcomes, and the research agenda in Section VIII sets out the empirical questions they raise.

Two features of this work are worth stating at the outset, since both are deliberate choices suited to its aim. It proceeds by theoretically-driven synthesis rather than a systematic-review protocol, integrating established traditions around a single problem for which relevance to the core tensions, not

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exhaustive coverage, is the appropriate selection principle; and it is a conceptual Perspective whose empirical validation is planned as follow-up work, in keeping with the usual sequence in which a framework is first articulated and then tested. What a paper of this kind owes the reader is a different, statable kind of rigor: not the statistical rigor of sample size or effect estimates, which it does not claim, but the discipline with which the argument is constructed and bounded. We therefore hold the framework to five explicit standards. First, a stated selection criterion: sources were included for their direct bearing on the core tensions the framework addresses rather than for exhaustive coverage. Second, traceability: each of the five dimensions and seven pillars is anchored in an established body of theory, named at the point of use, rather than asserted. Third, internal consistency: a single polarity lens is applied uniformly across every construct. Fourth, transparency about the boundary between what is claimed now, the integration and the account of how the parts interact, and what is deferred to future empirical work, namely the framework's effectiveness and the psychometric properties of its instruments. Fifth, falsifiability: the research agenda of Section VIII states the observations that would corroborate or disconfirm the framework's claims. The argument is, as in any Perspective, shaped by the author's judgment; these five standards are what hold that judgment accountable to evidence, and they are the sense in which a conceptual Perspective like this one earns its rigor.

B. Methodological Approach

Our methodology proceeds through four stages:

1) *Problem identification through interdisciplinary analysis.* We identified the inside-outside paradox by analyzing converging evidence from multiple research streams: empirical studies documenting cognitive debt from AI over-reliance [16], systematic reviews of AI's effects on cognitive abilities [17], meta-analyses of AI-scaffolded learning outcomes [18], [19], the automation literature on ironies of automation [34]–[36], and philosophical analyses of technology's effects on human capability [4], [37], [38]. The convergence of these streams reveals a persistent tension that cannot be resolved by choosing one pole, a hallmark of a polarity rather than a problem [9].
2) *Theoretical integration.* We synthesized polarity management theory, capability development philosophy, and writing pedagogy into a unified framework. The literature corpus drew on four theoretical streams: (a) empirical and experimental studies of AI's effects on cognition and writing (including meta-analyses, cognitive debt studies, and automation research); (b) classical and contemporary writing pedagogy (the canonical guides whose principles ground the CRAFTED pillars); (c) polarity management and paradox theory; and (d) capability development philosophy. Sources were selected through theoretically-driven synthesis rather than a systematic review protocol; the criterion was direct relevance to one or more of the core tensions the framework addresses. The five CRAFT dimensions were identified through iterative synthesis across the literature on writer formation, self-regulated learning [39], capability development [22]–[24], and skilled practice [37], [40], asking what qualities distinguish authentic authorship from sophisticated artifact production in AI-augmented contexts. They represent one defensible, integrative organization of these constructs rather than the only one possible, not a mechanical derivation from any single prior framework, and should be understood as theoretical constructs awaiting empirical scrutiny. Similarly, the seven CRAFTED pillars emerged from re-reading established writing pedagogy [27]–[29] through the lens of the inside-outside tension, identifying persistent polarities in each domain of writing craft. The specific counts (five dimensions and seven pillars) are a pragmatic, mnemonic design choice rather than a claim of exhaustiveness; finer or coarser groupings are possible. Each dimension and pillar is grounded in established theory, but the specific integration and the claims about how they interact constitute the novel contribution of this paper, one that requires empirical testing to be fully established.
3) *Framework articulation with exploratory self-assessment tools.* We operationalized the framework through polarity maps, developmental trajectories, and dual self-assessment instruments (evaluating both writer capability and artifact quality). These instruments are offered as exploratory, theoretically-grounded proposals that have not yet been empirically validated; they represent a first step toward operationalization intended to enable practitioners to engage with the framework immediately while formal instrument development and psychometric validation proceed as future work.
4) *Normative guideline development.* Drawing on the synthesized evidence base, we developed practical guidelines and ethical navigation principles. These are normative in character, prescribing how writers *should* navigate the paradox based on the best available evidence, consistent with the Perspective paper genre.

C. Scope and Limitations of this Approach

As a Perspective paper, our work has characteristic limitations. We do not report controlled experiments, survey data, or statistical analyses. The framework's effectiveness in improving writing outcomes or capability development remains to be empirically tested. The self-assessment tools, while grounded in established theory, require psychometric validation. The normative guidelines, while evidence-based, reflect our synthesis and judgment rather than experimentally established best practices. We address these limitations forthrightly in Section VIII and outline a detailed research agenda for future empirical work. We note that these limitations are inherent to the Perspective genre and do not diminish the contribution: a well-articulated, theoretically grounded framework that addresses an urgent practical challenge provides essential infrastructure for the empirical research program that must follow.

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III. Theoretical Foundations: Navigating the Inside-Outside Paradox

The CRAFT(-ED) framework addresses the inside-outside paradox through three integrated theoretical pillars. *Polarity management theory* explains why AI integration represents an ongoing tension rather than a problem to solve. *Multiplexity thinking* provides the both-and logic for navigating these tensions. *Capability-development philosophy* grounds why the inside pole matters at all, what the writer stands to become, or to lose, through AI use. Together, these three foundations support a practical framework for simultaneous attention to inside-focused capability development (CRAFT) and outside-focused effective communication (CRAFTED).

A. The Inside-Outside Paradox and Its Failure Modes

The *outside* dimension encompasses observable artifacts: the quality, coherence, and effectiveness of written outputs. It focuses on reader experience, communication effectiveness, and performance measured by conventional standards. AI excels at this dimension, producing professionally polished text with remarkable consistency. The *inside* dimension encompasses writer formation: the understanding, capability, authentic voice, and agency that develop through the writing process. It focuses on what the writer becomes through writing, the genuine comprehension achieved and the judgment cultivated. This dimension resists automation, emerging only through direct engagement with ideas and persistent struggle with expression. Pursued in isolation, each pole produces a characteristic failure, and neglecting both produces a third; the inside-outside paradox thus generates three distinct failure modes:

1) *AI Slop (Over-Emphasis on Outside)* Polished outputs lacking authentic understanding. Work appears impressive while masking shallow comprehension.
2) *Communication Failure (Over-Emphasis on Inside):* Genuine understanding that remains poorly expressed or inaccessible. Valuable insight effectively lost through a communication failure.
3) *Compound Mediocrity (Neglecting Both)* Neither AI's capabilities nor human understanding meaningfully developed, the highest opportunity cost.

The paradox matters profoundly for education because writing is not merely a communication skill but a formative practice. As Sennett [40] argues, craftsmanship represents "*an enduring, basic human impulse, the desire to do a job well for its own sake*" (p. 9). When AI short-circuits the developmental struggle that transforms writers, it enables production without formation. Yet completely rejecting AI perpetuates persistent weaknesses, unclear thinking, poor organization, inefficient processes, that limit authentic scholarship. Excellence requires navigating between these extremes.

B. Polarity Management: Why AI in Writing is a Polarity?

Barry Johnson's polarity management theory [9], [10] distinguishes between *problems to solve*, convergent challenges admitting permanent solutions, and *polarities to manage*, divergent challenges involving interdependent pairs of values that need each other over time. Attempting to "solve" a polarity by choosing one pole inevitably produces negative consequences demanding the opposite pole. AI integration in writing represents a polarity, not a problem. There is no "use AI" or "don't use AI" solution that works permanently across all contexts. Rather, writers must navigate ongoing tension between capability development and productivity, between authentic voice and effective assistance, between formation and output. As Schumacher [21] observed, some challenges are "divergent problems" requiring wisdom rather than technique, ongoing judgment rather than permanent answers. This diagnosis converges with recent assessment scholarship that characterizes AI-and-assessment as a wicked problem in Rittel and Webber's sense [41], [42]: one that cannot be solved once, only re-resolved continually. That organizational polarity theory and educational assessment research arrive independently at the same structural conclusion strengthens the case that AI integration is a polarity to be navigated rather than a problem to be closed.

Johnson's SMALL framework [10], [43] operationalizes polarity navigation through five iterative steps:

1) *Seeing:* Recognizing that AI integration is a polarity requiring ongoing navigation rather than a problem admitting permanent solution. The challenge is not "should we use AI?" but "how do we navigate between capability development and AI assistance?"
2) *Mapping:* Visualizing the upsides and downsides of each pole. CRAFT (inside) offers authenticity and deep understanding but risks inefficiency when overemphasized. CRAFTED (outside) offers efficiency and polish but risks superficiality when pursued alone. The map also identifies the Greater Purpose achievable through both poles and the Deepest Fear when both are mismanaged.
3) *Assessing:* Evaluating current reality through honest examination of which pole needs attention. Are writers over-relying on AI (losing inside development) or under-utilizing it (missing outside improvements)?
4) *Learning:* Developing indicators that signal when balance is shifting problematically. Early-warning signs include inability to explain AI-generated content (over-reliance) or persistent communication failures despite expertise (under-utilization).
5) *Leveraging:* Designing concrete action steps, structures, and practices to maximize both poles' upsides while minimizing downsides. This includes the specific guidance provided through each CRAFT dimension and CRAFTED pillar.

We apply this framework throughout CRAFT(-ED), providing polarity maps for each pillar that make visible the tensions writers must navigate. Following this methodology, we articulate what the inside-outside paradox enables at its best and threatens at its worst: *Greater Purpose:* "High performance with deep personhood", achieving excellent scholarly outputs while developing writers as thoughtful, capable individuals who can navigate complex intellectual challenges independently. This captures what education should accomplish: both qualification (capable performance) and subjectification (au-

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thentic personhood). *Deepest Fear:* "Slippy performance with shallow personhood", producing superficially impressive work that masks and accelerates erosion of deep understanding and authentic capability. This represents the convergence of AI slop and missed human potential: neither achieving genuine excellence nor developing authentic capability.

C. The CRAFT(-ED) Multiplex Framework

The solution to the inside-outside paradox lies in what we call *multiplex thinking*, maintaining both dimensions simultaneously rather than choosing between them [7], [8]. This approach draws on paradox theory's demonstration that intractable organizational challenges require both-and thinking rather than either-or choices [25], [26].

It is worth stating precisely what multiplexity adds, since it is easily read as a synonym for both-and thinking. As developed by Şentürk [7] and extended in our earlier work [8], multiplexity resolves into three dimensions, and CRAFT(-ED) draws on all three. *Horizontal multiplexity* is the both-and discipline of holding several valid perspectives on one plane, the polarity logic that refuses to collapse a divergent tension into a single pole; this is the dimension that paradox theory and polarity management already articulate. *Vertical multiplexity* adds what those theories leave implicit, namely an ordering of value across levels, so that competing goods are not merely balanced but ranked. In CRAFT(-ED) this ordering is decisive: the two poles are not weighted equally. We seek the efficiency and reach that outside-focused CRAFTED delivers, but we give higher priority to the inside-focused capability formation that CRAFT protects, a prioritization consonant with education's dual aim of qualification and subjectification [24], and it is this ordering that keeps the framework from sliding into mere productivity. *Dynamic multiplexity* adds a systems-and-process view: the balance between the poles is not fixed but evolves through feedback over time, and it is this dimension that the SMALL navigation cycle introduced above (Seeing, Mapping, Assessing, Learning, Leveraging) operationalizes, making the re-balancing continuous rather than a one-time settlement. It is the conjunction of these three, horizontal both-and reasoning, vertical value-ordering, and dynamic feedback, rather than any one alone, that distinguishes the multiplex reading of AI-augmented writing from a simple both-and framing, and that constitutes the theoretical contribution this Perspective claims. We keep the treatment deliberately compact, since multiplexity is elaborated at book length as a philosophical system elsewhere; our aim here is not to restate that system but to put its three dimensions to work on the specific problem of writing with AI.

Figure 2 illustrates the complete CRAFT(-ED) framework as a multiplex system integrating complementary perspectives in a dynamic relationship. The framework comprises three integrated components:

1) CRAFT (Inside-Focused Formation)

Five dimensions of writer development, **C**onsciousness (epistemic awareness of genuine understanding), **R**elational (authentic engagement with ideas and communities), **A**gentic (ownership and self-authorship), **F**ormative (capability development over time), and **T**ruthful (integrity and authenticity). This pole addresses: *Is the writer developing genuine understanding and irreplaceable capability?*

2) CRAFTED (Outside-Focused Communication)

Seven pillars ensuring output quality, **C**oherence (unified narrative), **R**eader focus (empathetic communication), **A**ccuracy (evidence-grounded claims), **F**low (seamless connectivity), **T**op-down thinking (strategic framing), **E**spirited authenticity (distinctive voice), and **D**irected design (intentional oversight). This pole addresses: *Is the work clear, accessible, and valuable to readers?*

3) Both-And Integration (The Navigation)

The infinity loop in Figure 2 represents how multiplexity is implemented between the two poles through the continuous navigation between poles, developing understanding, producing output, reflecting and verifying, refining capability. Neither pole suffices alone; wisdom lies in maintaining dynamic equilibrium as contexts shift, AI capabilities evolve, and individual capabilities develop. As the figure illustrates, CRAFT alone risks inefficiency and underutilizing technology's potential; CRAFTED alone risks superficiality and AI slop. Excellence emerges from leveraging both simultaneously, using AI to enhance efficiency and output quality *while simultaneously* developing authentic human capability. Not either-or, but both-and.

This multiplex perspective distinguishes CRAFT(-ED) from either pure skill-development approaches (which neglect output quality) or pure output-assessment approaches (which neglect capability formation). We refuse the false choice between "focus on learning process" or "focus on final product." Authentic scholarship requires both.

D. Philosophical Grounding: Capability and Formation

While polarity management provides the navigation structure, capability development philosophy explains what distinguishes genuine learning from mere artifact production. Three philosophical traditions ground the CRAFT dimensions:

Sen's capability approach [22] asks what people can do and be, not merely what they possess. Applied to writing: the question is not just whether students produce quality documents but whether they develop the capability to think clearly, express ideas effectively, and engage authentically with knowledge. Nussbaum's [23] elaboration identifies central capabilities including practical reason, affiliation, and control over one's environment, all essential to authentic authorship.

Biesta's educational philosophy [24], [44] distinguishes three purposes: *qualification* (developing skills), *socialization* (fitting into society), and *subjectification* (becoming an independent agent with unique voice and judgment). While AI can enhance qualification and complicate socialization, it most threatens subjectification, the formation of authentic selfhood through learning. Education, Biesta argues, involves becoming a responsible subject who can answer for their work, explain choices, and stand behind claims. This cannot be outsourced.

Crawford's [4] analysis of craftsmanship as "soulcraft" shows that skilled work fosters agency, attentiveness, and

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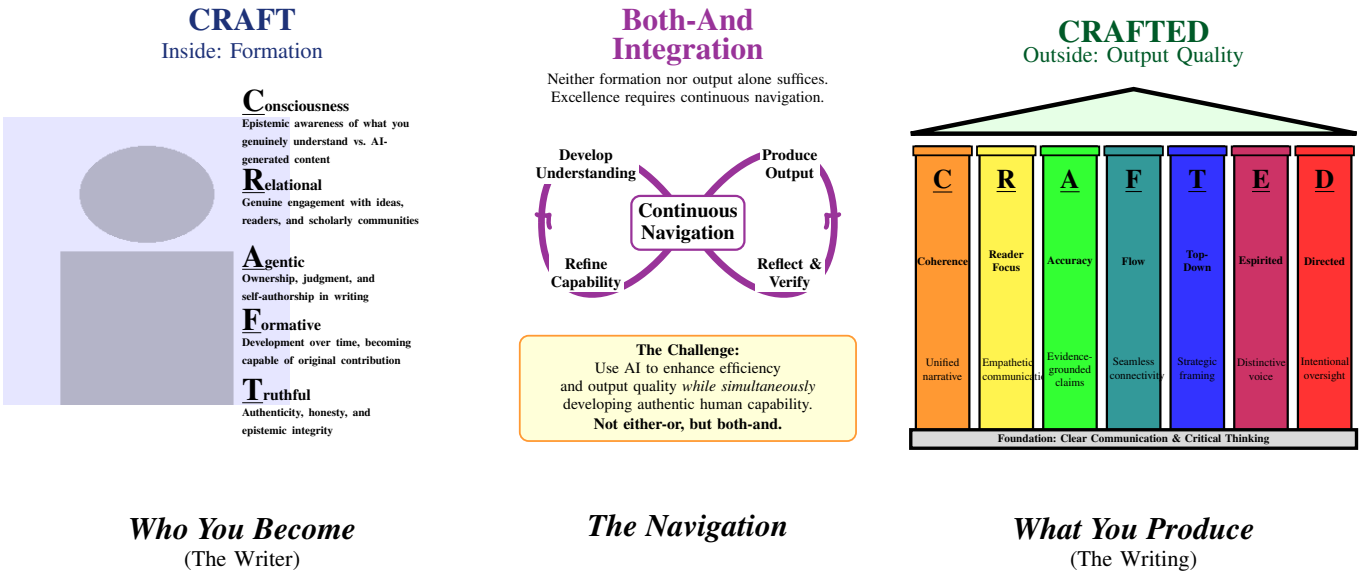


Fig. 2: **The CRAFT(-ED) Framework: Navigating the Inside-Outside Paradox.** Excellence requires continuous navigation between formation (CRAFT: **C**onsciousness, **R**elationality, **A**gency, **F**ormation, **T**ruthfulness) and output quality (CRAFTED: **C**oherence, **R**eader focus, **A**ccuracy, **F**low, **T**op-down, **E**spirited, **D**irected). The infinity loop represents both-and integration, either pole alone fails; continuous navigation enables authentic excellence.

cultivation of a self anchored in reality. Classic works on writing craft [27], [28], [45] make the same point: good writing is inseparable from the writer's thinking, judgment, and deliberate practice. Writing is craft because the writer must be formed: clarity grows from conceptual discipline, voice emerges from reflection, coherence arises from understanding. AI can provide polish and speed, but it cannot substitute the formative inner work that gives writing depth and integrity.

Having established theoretical foundations, we now turn to practical implementation. Section IV details the five CRAFT dimensions for inside-focused capability development. Section V presents the seven CRAFTED pillars for outside-focused effective communication. Each dimension and pillar explores the specific polarity it navigates, the connection between capability and communication, common pitfalls in AI-augmented writing, self-assessment tools for both dimensions, and concrete guidance for maintaining dynamic balance.

IV. CRAFT: Inside-Focused Capability Development

Having established that AI-augmented writing requires both inside-focused formation and outside-focused expression, we now detail the five CRAFT dimensions that constitute authentic writer development. The CRAFT framework articulates what Crawford [37] calls "skilled engagement with reality", the qualities of consciousness, relationship, agency, formation, and truth-commitment that distinguish genuine scholarship from sophisticated artifact production. These dimensions answer the question: *What does it mean to become a capable writer, not merely to produce capable writing?* They represent qualities that resist commodification and cannot be delegated to AI, dimensions of human formation that emerge only through direct engagement with ideas and persistent struggle with expression. While CRAFTED pillars (Section V) provide practical guidance for creating effective outputs, CRAFT dimensions address the parallel question of writer development. Both are essential; neither suffices alone.

The CRAFT dimensions engage with but extend established self-regulated learning (SRL) theory [39]. SRL frameworks describe the metacognitive processes, goal-setting, self-monitoring, adaptive regulation, by which learners control their own cognition. The Agentic and Formative dimensions draw substantially on these insights. However, AI-augmented writing introduces a structural challenge that SRL alone does not address: the possibility of apparent self-regulation that is actually AI-mediated regulation. We hypothesize that a writer who delegates planning, monitoring, and revision to AI may pass conventional SRL assessments while developing neither genuine understanding nor authentic authorial judgment, a risk warranting empirical investigation but one grounded in the broader automation and cognitive offloading literature [16], [34]. The CRAFT dimensions specifically target this gap, addressing what writers must cultivate *internally* to prevent the gradual substitution of AI processes for genuine human development.

Table I provides an overview of the five CRAFT dimensions, each addressing a different facet of authentic authorship in AI-augmented contexts. The dimensions operate as an integrated system: working on any CRAFTED pillar simultaneously develops multiple CRAFT dimensions. For instance, improving coherence (CRAFTED) requires consciousness of logical relationships, agency in organizational choices, and commitment to truthful representation of ideas.

1) *Consciousness:* Epistemic awareness distinguishing genuine understanding from AI-generated content
2) *Relational:* Engagement with ideas, readers, and scholarly communities
3) *Agentic:* Ownership, judgment, and self-authorship in the

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TABLE I: The Five CRAFT Dimensions: Inside-Focused Writer Formation

Dimension	*Core Focus*	*Key Question for Writers*	*Developmental Trajectory*
Consciousness	Epistemic awareness	Can you explain every major claim without referring to notes? Do you know what you genuinely understand versus what AI generated?	From accepting AI output uncritically to maintaining clear awareness of comprehension boundaries
Relational	Genuine engagement	Have you authentically engaged with key sources and ideas, or merely used AI summaries? Does your work contribute to ongoing scholarly conversations?	From extractive use of sources to participating in intellectual communities
Agentic	Self-authorship	Can you defend every major organizational and stylistic choice? Are you directing AI, or is AI directing you?	From passive acceptance of suggestions to full authorial ownership and strategic tool use
Formative	Capability development	Are you developing capabilities through this work, or merely assembling outputs? Would your abilities remain if AI disappeared?	From viewing writing as task completion to approaching it as formative practice
Truthful	Epistemic integrity	Can you stand behind every claim, explaining, defending, and taking responsibility? Have you been transparent about AI's role?	From unclear ethical boundaries to internalized epistemic responsibility

writing process
4) *Formative:* Development over time, becoming someone capable of original contribution
5) *Truthful:* Commitment to authenticity, honesty, and epistemic integrity

These dimensions are not sequential stages but complementary aspects of writer development. Working on any CRAFTED pillar simultaneously develops multiple CRAFT dimensions. For instance, improving coherence (CRAFTED) requires consciousness of logical relationships, agency in organizational choices, and truth-commitment to accurate representation.

A. Consciousness: Epistemic Awareness

Drawing on the Arabic concept of *shaur* (consciousness, awakening), this dimension emphasizes learning as the cultivation of awareness rather than mere information accumulation. As Rumi observed, "Knowledge that takes you not beyond yourself is far worse than ignorance." True learning transforms consciousness, how we see, understand, and make meaning.

In AI-augmented writing, consciousness means developing awareness of what you genuinely understand versus what AI has generated. It requires cultivating discernment about quality and appropriateness, deepening comprehension through engagement rather than outsourcing understanding to algorithms. When writers use AI without developing consciousness, they produce text they cannot explain, defend, or meaningfully revise. They possess artifacts without understanding, like copying not just answers but also the name at the top of the page.

Developmental Trajectory: Novice writers accept AI output uncritically, unable to distinguish between what they understand and what sounds plausible. Competent writers develop skepticism, checking AI-generated content against their understanding. Expert writers use AI as a tool for exploration while maintaining clear awareness of their own comprehension boundaries. They can articulate exactly what they know, what they've verified, and what remains uncertain.

AI Integration Implications: Consciousness cannot be delegated. It emerges only through direct engagement with ideas and persistent questioning of one's own comprehension. AI can provide information and suggest interpretations, but understanding, genuine epistemic awareness, requires human cognitive work. The automation paradox applies here: using AI effectively requires *more* sophisticated consciousness, not less, because you must evaluate outputs against your understanding.

Self-Assessment: Can you explain every major claim in your document without referring to notes? Could you restructure your argument if asked, maintaining its conceptual integrity? Do you know which parts reflect your genuine understanding versus AI elaboration you've accepted? If someone challenged your central thesis, could you defend it from first principles?

B. Relational: Engagement and Dialogue

Building on Flores and Winograd's [46] analysis of work as linguistic coordination and Wegerif's [47] dialogic theory of education, this dimension recognizes that learning occurs through relationships, with texts, ideas, peers, and mentors. Writing is not merely artifact production but participation in scholarly conversations, making commitments to readers, and engaging with intellectual communities.

AI-augmented writing risks reducing this relational dimension to mechanical production. The writer-reader relationship becomes transactional rather than dialogical; engagement with sources becomes extractive rather than conversational. Authentic scholarship requires maintaining the relational web, understanding how your work contributes to ongoing dialogues, honoring commitments to readers through clarity and accuracy, and engaging genuinely with ideas rather than merely assembling AI-generated summaries.

Developmental Trajectory: Novice writers see writing as individual production, focused on meeting requirements. They extract quotes from sources without genuine engagement.

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Competent writers recognize writing as communication, actively considering reader needs. Expert writers understand writing as participation in communities, contributing to conversations, building on others' work respectfully, and creating value for readers they may never meet.

AI Integration Implications: AI can help identify relevant sources and suggest connections, but authentic relational engagement requires human presence. You cannot truly engage with another scholar's ideas by reading AI's summary, you must wrestle with their arguments, understand their context, and place yourself in dialogue with their thinking. The relational dimension transforms writing from information transfer to intellectual conversation.

Self-Assessment: Can you articulate how your work contributes to existing scholarly conversations? Have you genuinely engaged with key sources, or merely used AI summaries? Do you consider specific readers' needs, or write generically? Would you feel comfortable discussing your topic with experts in the field?

C. Agentic: Ownership and Self-Authorship

Drawing on the Greek concept of *poiesis*, making that transforms the maker, this dimension emphasizes that writing develops the writer. The act of composing is not merely producing an artifact but cultivating judgment, taste, and voice. As Crawford [37] argues, skilled practice requires "embodied, situated, attentive engagement with reality" that cannot be outsourced without atrophy of capacity.

Agency in AI-augmented writing means making deliberate choices about what to delegate and what to craft directly, maintaining ownership of your document's purpose and direction, and developing through struggle rather than always choosing efficiency. The paradox, as Bainbridge [34] noted about automation, is that authentic AI use requires *more* sophisticated judgment, not less. To use AI well requires deeper domain understanding to evaluate outputs, stronger discernment to decide what to accept or reject, and deliberate practice developing capabilities AI doesn't provide.

Developmental Trajectory: Novice writers have limited agency, they follow templates and accept suggestions without evaluation. Competent writers exercise selective agency, they make conscious choices about structure, style, and content. Expert writers possess full authorial agency, they direct AI tools strategically, maintain ownership of all decisions, and can explain every choice based on principled reasoning.

AI Integration Implications: Agency distinguishes between using AI as tool versus being used by it. Writers with agency direct the process: they prompt strategically, evaluate outputs critically, and revise extensively. Writers lacking agency become passive assemblers, accepting AI suggestions without understanding or evaluation. The test: Are you directing AI, or is AI directing you?

Self-Assessment: Can you defend every major organizational and stylistic choice in your document? Do you maintain ownership of your work's direction, or drift wherever AI suggestions lead? Have you developed capabilities through this work, or merely assembled outputs? Could you produce quality work without AI assistance if required?

D. Formative: Development and Becoming

Biesta [24], [44] distinguishes three purposes of education: *qualification* (developing skills), *socialization* (fitting into society), and *subjectification* (becoming an independent agent with unique voice and judgment). While AI can enhance qualification and complicate socialization, it most threatens subjectification, the formation of authentic selfhood through learning.

In writing, formation means developing distinctive voice that reflects genuine perspective, cultivating judgment about quality and value, and becoming someone capable of original contribution rather than competent assembly. AI threatens to deliver the commodity, polished text, while bypassing the formative process that constitutes writers as practitioners. As Borgmann's [38] device paradigm illustrates, modern technology delivers outputs while withdrawing the focal practices that develop human capacity. AI writing risks this same pattern: delivering texts without developing writers.

Developmental Trajectory: Novice writers focus on producing acceptable outputs without recognizing the formative opportunity. They see writing as task completion. Competent writers begin recognizing how writing develops their thinking and capabilities. Expert writers approach writing as practice, understanding that the process forms them as much as they form the text. They deliberately use writing as a tool for developing understanding and capability.

AI Integration Implications: Formation requires sustained engagement over time. Quick AI-assisted production may deliver outputs but bypass the developmental struggle that transforms writers. The challenge is using AI as scaffold rather than replacement, leveraging assistance while maintaining the formative work that only struggle provides. This requires deliberate practice: regularly writing without AI, revising AI-generated content extensively, and consciously reflecting on your development.

Self-Assessment: Are you developing capabilities through AI-augmented writing, or merely maintaining access to AI-generated outputs? Can you identify specific ways your writing capability has grown? Do you approach writing as formative practice or merely output production? Would your capabilities remain if AI disappeared tomorrow?

E. Truthful: Authenticity and Integrity

The final dimension addresses commitment to truth, integrity in representation, and meaningful contribution. This goes beyond accuracy (covered in CRAFTED) to encompass epistemic responsibility, accountability for every claim made, and what Howard [48] calls "ethics as decision-making under uncertainty."

In AI contexts, truthfulness means verifying every AI-generated fact rather than trusting algorithmic confidence, maintaining transparency about AI's role in your work, and ensuring your contribution adds genuine value rather than merely repackaging existing material. Truthfulness also means honesty about understanding. Presenting AI-generated content you don't comprehend is a failure of epistemic integrity, regardless of the text's surface quality. This concern is formalized by Bakhshi [49], who shows that LLMs exhibit systematic

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epistemic-rhetorical miscalibration: they express stronger confidence than their actual knowledge warrants, producing text that sounds more certain than it has grounds to be. Writers who uncritically adopt such text inherit this miscalibration, undermining the truthfulness of their own scholarly voice.

Developmental Trajectory: Novice writers may not recognize the ethical dimensions of AI use, they see it as productivity tool without implications. Competent writers understand the need for verification and transparency but may struggle with consistent implementation. Expert writers have internalized epistemic responsibility, they automatically verify claims, maintain clear boundaries between AI and human contribution, and never present understanding they don't possess.

AI Integration Implications: Truthfulness creates nonnegotiable boundaries. You cannot ethically present AI-generated content as your own understanding. You cannot cite sources you haven't read. You cannot make claims you cannot explain. These boundaries don't make AI unusable, they define how to use it with integrity. The formative aspect: practicing truthfulness develops your capability for honest self-assessment and clear attribution.

Self-Assessment: Can you stand behind every claim in your document, explaining, defending, and taking responsibility? Have you been transparent about AI's role? Have you verified all factual content? Does your work reflect your authentic understanding and perspective? Would you feel comfortable presenting this work to experts in your field?

F. CRAFT Integration: How Dimensions Work Together

The five CRAFT dimensions are not isolated attributes but interconnected aspects of authentic development. Consciousness enables agency (you can't make good decisions without awareness). Relational engagement develops truthfulness (dialogue with others calibrates your understanding). Agency requires formation (capability developed over time). Truthfulness depends on consciousness (knowing what you actually understand).

In practice, working on any CRAFTED pillar simultaneously develops multiple CRAFT dimensions. Improving coherence (CRAFTED) requires consciousness of logical relationships (CRAFT: Consciousness), agency in organizational choices (CRAFT: Agentic), and commitment to truthful representation (CRAFT: Truthful). Enhancing reader focus (CRAFTED) develops relational awareness (CRAFT: Relational) and formative empathy (CRAFT: Formative).

This integration explains why CRAFT(-ED) operates as a multiplex system rather than a checklist. The dimensions and pillars are not separate items to check off but complementary perspectives on authentic AI-augmented scholarship. Section V details the seven CRAFTED pillars, each grounded in these CRAFT dimensions while providing specific practical guidance.

V. CRAFTED: Outside-Focused Effective Communication

While CRAFT dimensions address writer formation, the inside-focused question of becoming capable, CRAFTED pillars address effective expression, the outside-focused question of producing valuable work for readers. These seven pillars provide practical guidance for creating documents that are coherent, accessible, accurate, well-connected, strategically organized, engaging, and purposefully designed. Each pillar navigates a specific polarity tension inherent in AI-augmented writing.

The CRAFTED framework emerged from systematic analysis of what makes academic writing effective across disciplines, enhanced by recognition of AI's specific affordances and risks. While these principles apply to all writing, AI integration makes them simultaneously more accessible and more challenging: AI can help with structure and polish, but authentic effectiveness requires human judgment about audience, purpose, and quality.

Two points of framing are important before proceeding. First, the self-assessment instruments, polarity maps, and developmental trajectories presented throughout this section are *exploratory theoretical proposals* that have not been empirically validated. They are grounded in established theory and offered as provisional starting points for practitioners and future instrument development, not as ready-to-deploy tools with established reliability or validity. Their practical usefulness and educational impact remain to be determined through empirical research. Being unvalidated as new instruments does not, however, make them ungrounded: each element is derived from, and inherits support from, prior theory and evidence in its home domain, cited at the point of use, so the instruments are best read as evidence-based conceptual proposals rather than untethered speculation. Second, alongside the practical guidance that characterizes a Perspective paper, each pillar is grounded in the theoretical architecture established in Section III: the polarity management logic, the capability development philosophy, and the multiplex both-and approach. The practical tools are not substitutes for theoretical understanding but are intended as bridges between theory and everyday writing practice, consistent with IEEE Access's welcome of papers that integrate theoretical rigor with applied relevance.

A. The Seven-Pillar Architecture

Figure 2 visualized CRAFTED as seven pillars supporting excellence in AI-augmented writing. Like architectural pillars, each is necessary, remove one and the structure weakens. Yet they are not isolated; they work in concert, each strengthening the others when properly balanced. Readers are reminded that the self-assessment instruments embedded within each pillar are exploratory theoretical proposals, as noted in the section opening; they are not validated instruments but starting points for practitioner engagement and future empirical development. The seven pillars of CRAFTED are:

1) *Coherence:* Unified narrative structure across all levels
2) *Reader Focus:* Empathy, accessibility, and genuine service to readers
3) *Accuracy:* Groundedness in evidence and verifiable truth
4) *Flow:* Seamless connectivity between ideas
5) *Top-Down:* Strategic framing and big-picture perspective
6) *Espirited:* Human authenticity, voice, and engagement
7) *Directed:* Intentional design and purposeful oversight

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TABLE II: The Seven CRAFTED Pillars: Outside-Focused Communication Quality

Pillar	*Core Focus*	*Key Question for Writers*	*Primary CRAFT Connection*
Coherence	Unified narrative	Does your document tell a single, integrated story, or does it jump between disconnected ideas?	Consciousness (understanding logical relationships)
Reader Focus	Empathetic communication	Have you genuinely considered your readers' needs, background, and constraints?	Relational (engagement with audience)
Accuracy	Evidence-grounded claims	Have you personally verified every factual claim, statistic, and citation?	Consciousness + Truthful (verification and integrity)
Flow	Seamless connectivity	Do transitions reflect actual logical connections you understand, or just AI-generated bridges?	Consciousness + Agentic (understanding and deliberate choice)
Top-Down	Strategic framing	Can you state your core message in one sentence and explain how each section serves it?	Agentic + Formative (strategic ownership)
Espirited	Distinctive voice	Does this writing sound authentically like you, or could it have been written by anyone?	Agentic + Formative (self-authorship and voice)
Directed	Intentional oversight	Are you directing AI as a tool, or passively accepting whatever it generates?	Agentic (meta-cognitive control)

Table II provides an overview of the seven CRAFTED pillars with their connections to CRAFT dimensions. As the table illustrates, improving output quality is inseparable from developing writer capability. You cannot achieve genuine coherence without consciousness of logical relationships; you cannot demonstrate authentic reader focus without relational engagement; you cannot maintain accuracy without truthful commitment to verification. The pillars rest on a foundation of clear communication and critical thinking, capabilities that AI can assist but never replace.

Each of the pillar subsections below follows a consistent structure: definition and importance, the underlying polarity, connection to CRAFT dimensions, tips for enhancement, common pitfalls, and dual self-assessment (inside capability and outside artifact quality). This architecture is designed to support systematic navigation of AI-augmented writing challenges.

B. Pillar 1: Coherence, Unified Narrative Structure

Coherence is the quality that makes a document feel like a unified whole rather than a collection of disconnected parts. It operates at multiple levels: between sentences within paragraphs, between paragraphs within sections, between sections within the document, and between the document's parts and its overarching purpose. In effective writing, each element connects logically to what precedes and follows it, creating seamless narrative flow that guides readers through the argument.

In the GenAI era, coherence becomes both easier and harder. AI models excel at local coherence, generating grammatically correct sentences that connect plausibly to adjacent content. However, this surface-level coherence can mask deeper structural incoherence. A document may read smoothly while lacking genuine logical unity, a phenomenon we might call "counterfeit coherence."

1) The Polarity: AI-Suggested Structure ↔ Human Understanding

Coherence navigates the tension between AI's structural suggestions and the writer's own understanding of logical relationships. AI-suggested structure helps overcome blank-page paralysis and offers templates, transitions, and gap-spotting, but over-reliance imposes generic patterns, manufactures plausible-but-superficial connections, and tempts writers to accept structure they do not understand. Human understanding secures genuine conceptual grasp, adaptive restructuring, and the ability to defend one's choices, but leaned on alone it is slower and can yield idiosyncratic structures that confuse readers or miss better AI-suggested alternatives.

The navigation: Use AI for structural suggestions, but never accept organization you don't understand. Transform AI's scaffolding into your own conceptual framework.

2) CRAFT Connection

Coherence exercises several CRAFT capabilities at once: grasping how ideas relate (Consciousness), building structural judgment by working out why an AI-suggested structure does or does not hold (Formative), respecting readers' cognitive effort (Relational), and owning organizational choices so the voice stays yours even when AI assists (Agentic, Truthful).

The test is simple: Can you explain why each section follows the previous one? Can you create an outline from your draft that reveals its logic? If AI restructured your document, could you evaluate whether the changes preserve or damage its conceptual integrity?

3) Tips for Enhancing Coherence

Creating logical links. Effective writing establishes strong logical connections between ideas and sections. Use signposting phrases to guide readers through your argument's logic. Transition words such as "however," "therefore," and "moreover" can clarify relationships between ideas, ensuring readers follow your reasoning. However, ensure transitions

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reflect genuine conceptual relationships, not just grammatical convenience.

Theme and variations. Identify your central theme and explore its variations. This approach provides depth while maintaining consistency. Each section should echo the main theme, offering new perspectives that contribute to the overall narrative. AI can help identify themes, but you must ensure thematic unity reflects actual conceptual relationships.

Outline before, during, and after. Create outlines before writing to visualize structure. Revisit and adjust outlines during drafting to maintain coherence. After completing a draft, create a reverse outline, write one-sentence summaries of each paragraph. This reveals whether your document has clear hierarchical structure or is merely well-written paragraphs without authentic organization.

Using AI for structure, not understanding. When using AI to suggest organizational structures, treat suggestions as starting points for understanding, not endpoints. Ask yourself: Why does this structure work? How does it serve my argument? What would change if I reorganized differently? This transforms AI from a crutch to a learning scaffold.

The shuffle test. Print paragraphs on separate pages and shuffle them. Can a reader reconstruct the correct order based on logical flow? If not, your transitions and signposting need strengthening. AI can help identify weak links, but you must understand and fix them.

4) Common Pitfalls

The most dangerous pitfall: accepting an AI-suggested structure without grasping the underlying logic. This produces documents that read smoothly but lack authentic conceptual unity. Other common failures include writing distinct sections in isolation without a clear sense of their interconnection (fragmented writing), and excessive compartmentalization with too many subsections that fragment narrative flow.

5) Coherence Self-Assessment

The following boxes provide manual and AI-assisted **Coherence** audit tools:

CRAFT Assessment: Coherence

Test your understanding:

(a) Can you explain why each major section belongs in your document and why it appears in this particular order?
(b) Could you restructure your document if asked, maintaining its logical integrity?
(c) If someone questioned your structure, could you defend it based on logic rather than "AI suggested it"?

CRAFTED Assessment: Coherence

Evaluate your output:

(a) Does each section follow logically from the previous one?
(b) Are transitions smooth and meaningful, not just grammatically correct?
(c) Is there a clear narrative arc from introduction to conclusion?

AI-assisted audits can support but never replace human judgment. The following prompt helps identify potential coherence issues.

DIY: AI-Assisted Coherence Self-Audit

Use this prompt with your favorite GenAI tool:
"Analyze this document for structural coherence. Identify: (1) any sections that seem disconnected from surrounding content, (2) weak or missing transitions between major ideas, (3) organizational choices that may confuse readers, (4) places where the logic behind the structure is unclear. For each issue, explain specifically WHY it matters for coherence."

Reflect: Do I understand the issues AI identified? Can I explain and fix them? Or am I simply accepting AI's next suggestion without genuine comprehension?

C. Pillar 2: Reader Focus, Empathy and Accessibility

Reader focus embodies empathy and respect for your audience. It requires writers to step outside their expert perspective and consider: Who will read this? What do they already know? What do they need to know? How can I make complex ideas accessible without oversimplification? This pillar transforms writing from a self-centered exercise into service, communication designed to genuinely help readers understand.

In AI-augmented contexts, reader focus becomes more complex. AI can help polish language and suggest clarifications, but it cannot substitute for a genuine understanding of your specific audience's needs, background, and interests. The danger is producing text that is technically clear but emotionally disconnected, grammatically correct prose that fails to connect with actual human readers.

1) The Polarity: AI Polish ↔ Human Empathy

Reader focus navigates the tension between AI's ability to polish language and the human capacity for authentic empathy. AI polish improves clarity and readability, flags jargon, aids multilingual writers, and adds accessibility, but over-reliance yields a generic tone that strips warmth, precision, and authentic voice and mistakes "readable" for "reader-focused." Human empathy brings genuine understanding of the audience, anticipation of their questions, and trust, but leaned on alone it can assume context readers lack, risk condescension, or miss real clarity gains AI could provide.

The navigation: Use AI for polish and accessibility checking, but ground all choices in a genuine understanding of your specific readers' needs and perspectives.

2) CRAFT Connection

Reader focus develops epistemic and ethical responsibility: you owe readers genuine understanding, not AI-generated filler (Relational, Truthful). Paradoxically it strengthens authentic voice, since personality shows through precisely when you want to help readers understand (Truthful). It also demands meta-cognitive judgment about how words will land for different audiences and when technical precision serves readers rather than raising barriers.

3) Tips for Enhancing Reader Focus

The over-the-shoulder technique. As you write, imagine a specific reader looking over your shoulder, perhaps a colleague from a related but different field, or a graduate student new to your area. Would they understand each sentence? Would they see why this matters? This technique prevents the curse of knowledge, experts often overestimate what readers know,

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assuming familiarity with concepts or context that may be unfamiliar.

Defining for your audience. Definitions should meet readers where they are. AI can suggest definitions, but you must judge whether they match your audience's background. Consider providing multiple levels of detail: brief in-text definitions for quick reference, deeper explanations in footnotes or appendices for those who need them.

Using examples and analogies. Concrete examples and apt analogies make abstract concepts tangible. When using AI to generate examples, ensure they genuinely illuminate rather than distract. The best examples connect to readers' existing knowledge and experiences.

Active voice for engagement. While AI often defaults to passive constructions, active voice typically engages readers more effectively. Make conscious choices based on what serves readers, not what AI defaults to. Compare "It was found by the researchers" (passive, distancing) with "Researchers found" (active, engaging).

AI as accessibility auditor. Use AI specifically to check accessibility: Are paragraphs digestible? Is jargon necessary? Would non-experts understand? But remember: AI can flag issues, but only you can judge how to address them for your specific audience.

Reader personas. Create detailed personas for your target readers. What do they care about? What confuses them? What motivates them to keep reading? Use these personas to evaluate both your writing and AI suggestions.

4) Common Pitfalls

AI-polished but soulless text is grammatically perfect yet emotionally flat. Reader focus requires warmth, personality, and human connection that purely AI-polished text often lacks. Other failures include neglecting audience needs (writing either too technical or too simplistic), and isolated literature reviews that discuss papers individually without connecting them to the broader context or each other, frustrating readers trying to understand the field.

5) Reader Focus Self-Assessment

The following boxes provide manual and AI-assisted **Reader Focus** audit tools:

CRAFT Assessment: Reader Focus

Test your empathy:

(a) Can you describe your target reader specifically? (Not "experts" but "civil engineering faculty with limited AI background")
(b) Have you anticipated and addressed likely reader questions?
(c) Could you explain your work to three different audiences, adjusting language and emphasis appropriately?

CRAFTED Assessment: Reader Focus

Evaluate your output:

(a) Would your target reader understand each section without extensive background?
(b) Is jargon used only when necessary, with clear explanation?
(c) Does the tone strike the right balance between authority and accessibility?

DIY: AI-Assisted Reader Focus Audit

Use this prompt with your favorite GenAI tool:
"Analyze this document from the perspective of [describe your target reader]. Identify: (1) terms or concepts that need more explanation, (2) sections that assume too much prior knowledge, (3) places where examples would help, (4) jargon that could be simplified without losing precision. For each issue, suggest specific improvements."

Reflect: Do I agree with these assessments based on my understanding of my actual readers? Are the suggestions genuinely reader-focused, or merely generic clarity advice?

D. Pillar 3: Accuracy, Groundedness in Evidence

In the GenAI era, accuracy takes on heightened significance. While AI models produce grammatically correct and coherent text, this proficiency does not guarantee factual correctness. The phenomenon of AI "hallucination", plausible but factually incorrect information, makes groundedness critical. Accuracy requires that content align with real-world evidence, that claims be verifiable, and that sources be correctly cited and represented.

Groundedness distinguishes genuine scholarship from sophisticated fabrication. The scale of this challenge is now empirically documented: Topaz et al. [50] audited 2.5 million biomedical papers and found that fabricated citations, references whose claimed titles correspond to no existing publication, are embedded in the peer-reviewed literature at scale, with the fabrication rate increasing more than twelvefold between 2023 and early 2026. Many of these fabricated references are attributable to uncritical use of AI writing tools, which generate plausible-sounding but fictitious citations that are difficult to detect by conventional peer review. As Bergstrom and West [51] emphasize, we must cultivate skepticism toward claims, especially AI-generated ones. Yet accuracy is not merely about fact-checking, it involves understanding the quality of evidence, recognizing the limits of knowledge, and representing uncertainty honestly.

1) The Polarity: AI Efficiency ↔ Human Verification

Accuracy navigates the tension between AI's rapid synthesis of information and the human responsibility for verification. AI efficiency speeds literature search, pattern-finding, source identification, and initial fact-checking, but over-reliance produces hallucinated citations, plausible-but-wrong claims, misrepresented arguments, and false confidence from polished presentation. Human verification supplies domain judgment of plausibility, source quality, and disciplinary standards, and ownership of every claim, but leaned on alone it is slow, can miss AI-accessible sources, and risks confirmation bias toward familiar work.

The navigation: Use AI for efficiency in finding sources, but verify everything personally. Never trust AI as source of factual information.

2) CRAFT Connection

Accuracy develops domain knowledge and critical evaluation, since you cannot verify claims you do not understand (Formative, Consciousness); checking AI-generated content against sources deepens that knowledge. At bottom it is epistemic responsibility, an accountability for what you claim

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that cannot be delegated (Agentic, Truthful); as Biesta [24] argues, responsibility is what makes us subjects rather than objects of education. Mature accuracy is knowing when to trust AI, when to verify, and what evidence each claim requires.

3) Tips for Enhancing Accuracy

The zero-shot translation approach. As Mittelstadt et al. [52] recommend, use LLMs as zero-shot translators, not information sources. Provide AI with verified facts and have it transform them into desired formats, never trust AI as the source of factual information.

Verify every citation. Check that cited sources exist, that they actually make the claims attributed to them, and that they are correctly formatted. This is non-negotiable. Even AI-generated citations that look plausible may be completely fabricated.

Sanity checks. Develop a skeptical eye for implausible claims. If AI generates a statistic, date, or technical detail that seems surprising, verify it independently. As Bergstrom and West [51] note, developing a strong bullshit detector is essential in the information age.

Primary sources when possible. Rather than relying on AI summaries or secondary sources, access primary sources directly. Company reports, peer-reviewed papers, and official statistics are more reliable than AI's interpretation of them.

Document your process. Maintain clear records of which content is AI-generated versus human-created, which sources you've personally verified, and what remains to be checked. This transparency serves both you and your readers.

Triangulation. Cross-reference important claims across multiple independent sources. If only one source supports a key claim, treat it with appropriate caution.

4) Common Pitfalls

Trusting AI-generated "facts" is the most dangerous pitfall. AI models are not knowledge bases; they are pattern-matching systems that can confidently generate plausible falsehoods. Other failures include citing sources AI suggested without reading them yourself (academically dishonest and practically dangerous), confirmation bias in verification (confirming what fits expectations while missing what doesn't), and letting polished presentation mask inaccuracy (grammatically perfect text does not equal factual accuracy).

5) Accuracy Self-Assessment

The following boxes provide manual and AI-assisted **Accuracy** audit tools:

CRAFT Assessment: Accuracy

Test your understanding:

(a) Can you explain & defend every factual claim in your document?
(b) Have you personally verified all citations and quotations?
(c) Could you distinguish which claims are well-supported versus speculative?

CRAFTED Assessment: Accuracy

Evaluate your output:

(a) Are all claims supported by credible, verifiable sources?
(b) Are citations accurate, properly formatted, and accessible?
(c) Is uncertainty acknowledged where appropriate?

The following audit identifies potential accuracy concerns, but verification remains your responsibility.

DIY: AI-Assisted Accuracy Audit

Use this prompt with your favorite GenAI tool:
"Review this document for factual accuracy concerns. Identify: (1) claims that seem implausible or need verification, (2) statistical or numerical claims that should be sourced, (3) citations that may be incorrect or incomplete, (4) places where evidence seems insufficient for claims made. Flag anything that raises accuracy concerns."

Critical next step: Do *not* trust AI's accuracy judgments blindly. Use AI to *flag* possible issues, then independently verify each one. The irony is using AI to check AI-generated content, but human judgment remains essential.

E. Pillar 4: Flow, Seamless Connectivity

Flow is the quality that allows readers to move effortlessly through a document, each idea naturally leading to the next. It operates at the sentence level (within-paragraph flow), paragraph level (within-section flow), and document level (overall progression). Good flow feels invisible, readers absorb ideas without noticing the transitions that carry them forward.

In AI-augmented writing, flow becomes paradoxical. AI excels at local flow, generating sentences that connect grammatically to their neighbors. Yet this can create what we might call "shallow flow", text that reads smoothly without genuine logical progression. Deep flow requires understanding the conceptual relationships that transitions reflect.

1) The Polarity: AI Connection ↔ Human Integration

Flow navigates between AI's capacity for smooth connections and the human understanding of logical integration. AI connection suggests effective transitions, flags abrupt jumps, and keeps grammar and tense consistent, but over-reliance manufactures surface fluency that masks conceptual gaps with generic transitions the writer accepts without understanding. Human integration ensures transitions reflect real logical relationships and can explain why ideas connect, but leaned on alone it can produce awkward transitions, skip conventional connective devices, or assume relationships that are not obvious to readers.

The navigation: Use AI to suggest transitions, but ensure they reflect genuine conceptual relationships you understand.

2) CRAFT Connection

Genuine flow reflects deep comprehension of how ideas relate, not surface connection; it develops your ability to perceive conceptual progression and logical relationships, which cannot be outsourced to AI. Even when using AI-suggested transitions, making them authentically represent your reasoning maintains intellectual integrity.

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content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3718869

3) Tips for Enhancing Flow

Signposting. Clear signposting guides readers through your argument's logic. Phrases like "Building on this foundation," "In contrast," or "This raises the question" explicitly mark logical relationships. AI can suggest these, but you must ensure they accurately reflect the actual connections.

Thematic unity. Each paragraph should have one main idea, and each sentence should contribute to that idea. AI can help identify paragraphs that lack unity, but you must understand what unifies them.

The last sentence–first sentence test. The last sentence of each paragraph should connect meaningfully to the first sentence of the next. Read these sentence pairs in isolation, do they reveal logical progression?

Varied sentence structure. While AI often generates varied structures, ensure variety serves readability rather than just creating surface variety. Sometimes simple, parallel structures enhance flow by making relationships clear.

The read aloud test. Reading your document aloud (especially transitions) helps identify where flow breaks down. Awkward reading usually signals conceptual gaps that smooth AI-generated transitions may mask.

4) Common Pitfalls

The most insidious pitfall: grammatical smoothness without logical depth. Text reads smoothly but lacks genuine conceptual progression. AI can create this; only human understanding can detect and fix it. Other failures include abrupt sectional transitions (smooth sentences within paragraphs but jarring jumps between sections, indicating structural problems that surface-level editing cannot fix), over-reliance on formulaic transitions (especially AI's favorites like "furthermore," "moreover," "additionally," creating mechanical rather than organic flow), and ignoring narrative arc (documents should have shape, building toward key points, providing necessary context before complex arguments).

5) Flow Self-Assessment

The following boxes provide manual and AI-assisted **Flow** audit tools:

CRAFT Assessment: Flow

Test your understanding:

(a) Can you explain how each paragraph connects logically to surrounding paragraphs?
(b) Do your transitions reflect genuine conceptual relationships, not just grammatical convenience?
(c) Do you understand the narrative arc of your document?

CRAFTED Assessment: Flow

Evaluate your output:

(a) Can readers follow your argument without getting lost?
(b) Do transitions feel natural and meaningful, not mechanical?
(c) Would readers experience the document as unified rather than assembled?

AI can help identify flow issues, but understanding and fixing them requires human comprehension of logical relationships.

DIY: AI-Assisted Flow Audit

Use this prompt with your favorite GenAI tool:
"Analyze this document's flow and transitions. Identify: (1) abrupt jumps between paragraphs or sections, (2) weak or missing transitions, (3) places where connections seem surface-level rather than logically deep, (4) inconsistencies in narrative progression. For each issue, explain what logical relationship is unclear or missing."

Then ask: Do I understand the logical gaps AI identified? Can I articulate what conceptual relationship is missing? Or am I simply relying on AI to fill gaps I myself do not yet understand?

F. Pillar 5: Top-Down Thinking, Strategic Framing

Top-down thinking means maintaining awareness of the big picture, the overarching purpose, argument, and structure, while working on details. It's the ability to zoom out from sentence-level editing to see how each part serves the whole. This pillar draws on Barbara Minto's Pyramid Principle [29], which emphasizes starting with the main message and supporting it with hierarchically organized details.

With GenAI, top-down thinking can become challenging as AI typically operates bottom-up, generating content without understanding global structure or purpose. In what has been termed the *"irony of automation"*, writers can easily get lost in polishing AI-generated details while losing sight of whether those details serve the document's core argument [36].

1) The Polarity: AI Details ↔ Human Framing

Top-down thinking navigates between AI's strength in elaborating details and the human responsibility for coherent framing. AI detail fleshes out outline points, supplies examples and supporting arguments, and fills gaps, but over-reliance yields detail that does not serve the main argument, local coherence without global purpose, and scope creep that blurs the core message. Human framing keeps a clear sense of purpose, makes strategic include/exclude choices, and supplies the "so what?", but leaned on alone it can stay too abstract, underdevelop arguments, or forgo useful AI elaboration. *The navigation*: Establish your core message and structure first, then use AI to elaborate within that framework, not vice versa.

2) CRAFT Connection

Top-down thinking develops responsibility for the document's purpose: you must keep asking why each element is included and how it serves readers and argument, and this cannot be delegated (Agentic). Holding the big picture while managing details exemplifies the meta-cognitive awareness of the Consciousness dimension, the oversight that keeps all parts cohering toward a purpose, and it is what lets flow be authentic rather than merely grammatical.

3) Tips for Top-Down Thinking

Start with the core message. Before generating any content, articulate your core message in one sentence. Every section should support this message. When AI generates content, evaluate it against this criterion: Does this serve my core message?

The Pyramid Principle. Following Minto [29], organize hierarchically: main point at top, supporting points below, details beneath those. Use AI to elaborate within this structure, not to create the structure itself.

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The reverse outline technique. After drafting (whether with AI or not), create a reverse outline: write a one-sentence summary of each paragraph. This reveals whether your document has clear hierarchical structure or is merely a collection of well-written paragraphs.

Literature review as synthesis. A top-down literature review groups papers thematically and shows relationships, not just summarizes each study individually. Use Polya's strategies [53]: identify patterns, find gaps, analyze relationships between works, trace methodological lineages.

Visual mapping. Create visual maps of your argument structure. Mind maps, concept maps, or simple hierarchical outlines help maintain big-picture awareness that AI-assisted drafting can obscure.

The elevator pitch test. If you can't explain your document's core argument in 30 seconds, you haven't clarified your top-down structure. This test should be performed before, during, and after AI-assisted drafting.

4) Common Pitfalls

Getting lost in AI-generated details is the most common pitfall: letting AI elaborate on tangential points while the core argument remains underdeveloped. Constantly return to: Does this serve my main purpose? Other failures include *isolated literature reviews* (discussing papers individually without connecting them to your argument or each other, reviews should be thematic and integrated, not bibliographic catalogs); *lack of chunking* (failing to organize content into meaningful categories and hierarchies makes work inaccessible), and *missing the "so what?"* (even well-organized content fails if it doesn't clearly convey why it matters).

5) Top-Down Thinking Self-Assessment

Manual and AI-assisted **Top-Down Thinking** audit tools are provided next:

CRAFT Assessment: Top-Down Understanding

Test your big-picture awareness:

(a) Can you state your document's core message in a clear sentence?
(b) Can you explain how each section contributes to this message?
(c) Do you know which parts are essential versus tangential?

CRAFTED Assessment: Top-Down Thinking

Evaluate your output:

(a) Is the main argument clear from the outset?
(b) Would readers understand the "so what?" of your work?
(c) Is there clear hierarchical organization, not just sequential content?

DIY: AI-Assisted Top-Down Audit

Use this prompt with your favorite GenAI tool:
"Create a hierarchical outline of this document showing: (1) the main argument or purpose, (2) how each major section supports the main argument, (3) key subsections and their roles, (4) any sections that seem tangential or poorly connected to the main purpose. Then evaluate: Does this document have clear top-down structure, or is it just well-written parts without strong organization?"

Reflect: Does this outline reveal the structure I intended? Do I agree with AI's assessment of which parts seem tangential? Can I defend every section's inclusion based on how it serves my core argument?

G. Pillar 6: Espirited, Human Authenticity and Style

> *"In literature the ambition of the novice is to acquire the literary language; the struggle of the adept is to get rid of it."*, G. B. Shaw

We coin *Espirited* from the French *esprit*, a term whose semantic range, encompassing spirit, wit, liveliness, and intellectual verve, has no single English equivalent. The Oxford English Dictionary traces *esprit* in English usage to the 1590s, noting its distinctive connotation of "sprightly wit or intelligence" beyond what *spirited* alone conveys. English has long borrowed the term in compounds such as *esprit de corps* (collective spirit) and *bel esprit* (a person of fine intellect). We adopt *Espirited* deliberately because existing alternatives fail to capture the precise quality this pillar demands: *spirited* emphasizes energy but lacks the intellectual dimension; *authentic* emphasizes truthfulness but not vitality; *engaging* emphasizes reader response but not writer voice. *Espirited* writing combines all three, lively human vitality, intellectual presence, and distinctive voice, qualities that resist the homogenization pressure of AI-generated text, which tends toward competent blandness. Espirited writing has personality, style, and authentic human presence that make work memorable and engaging beyond mere technical correctness.

This is perhaps the most fundamentally human pillar. AI can produce clear prose, but authentic voice, the qualities that make writing distinctively yours, emerges from personality, values, experience, and creative choice. The challenge is maintaining this authenticity while benefiting from AI's capabilities.

1) The Polarity: AI Formulaic Patterns ↔ Human Creativity

Espirited writing navigates between AI's reliable but generic patterns and human creative expression. AI patterns provide professional-sounding prose, overcome writer's block, and help non-native speakers write fluently, but over-reliance produces a voice indistinguishable from other AI-assisted work, homogenized across authors and predictable in phrasing. Human creativity brings distinctive voice, memorable metaphor, and personality that engages, but leaned on alone it risks idiosyncrasy that confuses readers, violates disciplinary norms, or becomes self-indulgent rather than reader-focused.

The navigation: Use AI for initial drafts, then revise extensively for voice and personality. Make the language yours.

2) CRAFT Connection

This pillar embodies the CRAFT Truthful dimension (authenticity and integrity), developing and maintaining a distinctive

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voice that reflects who you are as a thinker. Mature style is knowing when to follow conventions and when to break them: as Shaw notes, novices acquire literary language while experts liberate themselves from it. Authentic style is not self-indulgence but service, expressing personality in ways that enhance rather than obstruct communication.

3) Tips for Cultivating Espirited Writing

Style manuals as foundation. Strunk and White's *Elements of Style* [45], Pinker's *Sense of Style* [27], and Zinsser's *On Writing Well* [28] provide foundational guidance. Master the principles before transcending them.

The revision-for-style process. Revise AI-augmented text carefully, specifically for voice and personality. Remove generic AI phrases. Add metaphors, examples, and phrasings that reflect your perspective. Make the language yours.

Telling stories. As the Heath brothers [54] demonstrate, stories make ideas sticky. Integrating personal anecdotes, concrete examples, and narrative structures transforms informative writing into engaging communication. Use AI to help structure stories, not to generate them.

Consistency in voice. Maintain thematic unity and consistent voice from title through conclusion. AI often generates different tones in different sections, your job is to ensure a unified voice throughout.

Strategic use of first person. While conventions vary by discipline, strategic use of "I" or "we" can strengthen rather than weaken academic writing by clarifying agency and perspective. Don't let AI's preference for passive constructions eliminate your voice.

The personality test. Could a colleague who knows your work recognize this as yours based on style alone? If AI has completely effaced your personality, revision is needed.

4) Common Pitfalls

The most common pitfall is accepting AI's generic voice, competent but unmemorable. Other failures include prioritizing style over substance (clever phrasing that obscures meaning), treating first drafts as final (use multi-shot prompting to explore perspectives while maintaining human oversight), and sacrificing personal voice for technical soundness (your unique perspective adds irreplaceable value). AI can identify generic sections, but making work genuinely yours requires your voice and revision.

5) Espirited Writing Self-Assessment

The following boxes provide manual and AI-assisted **Espirited Writing** audit tools:

CRAFT Assessment: Espirited Authentic Style

Test your authentic voice:

(a) Does this writing sound like you, or like generic academic prose?
(b) Have you revised AI-generated text to reflect your perspective?
(c) Does your personality and passion for the topic come through?

CRAFTED Assessment: Espirited Authentic Style

Evaluate your output:

(a) Does voice remain consistent throughout the document?
(b) Are there moments of distinctive phrasing or insight?
(c) Would readers experience this as human-written, not AI-generated?

DIY: AI-Assisted Espirited Voice Audit

Use this prompt with your favorite GenAI tool:
"Analyze this document for voice and style. Identify: (1) sections that sound generic or formulaic, (2) overuse of common AI phrases (e.g., 'delve into,' 'tapestry,' 'landscape'), (3) passages that lack personality or distinctive perspective, (4) opportunities to make writing more engaging while maintaining professional tone."

Revise: Use AI's feedback to *locate* bland sections, but revise the prose yourself to reflect your authentic voice. Avoid asking AI to "make it more engaging", that often produces a different set of generic patterns. Make it genuinely yours.

H. Pillar 7: Directed, Intentional Design and Oversight

Directed refers to intentional, purposeful choices about document design, organization, and presentation. It encompasses visual design (figures, tables, formatting), organizational design (sectioning, headings, flow), and the meta-cognitive oversight that ensures all elements serve the document's purpose. In AI-augmented contexts, this pillar emphasizes human agency, you direct the AI, not vice versa.

This pillar addresses a fundamental issue in AI-augmented work: the distinction between mere assembly and genuine authorship. Directed writing requires active oversight, strategic choices, and clear ownership of the document's design and execution.

1) The Polarity: AI Automation ↔ Human Oversight

Directed writing navigates between AI's capacity for automated production and the human responsibility for intentional oversight. AI automation rapidly generates content, keeps style consistent, and handles standard elements efficiently, but over-reliance breeds passive acceptance, loss of authorial agency, generic design, and blurred human-AI boundaries. Human oversight supplies intentional, purpose-serving choices, clear ownership, and the judgment to reject AI suggestions, but leaned on alone it is time-intensive, forgoes AI's efficiency on routine tasks, and can over-control details that do not matter.

The navigation: Direct AI strategically for efficiency, but maintain active oversight and intentional decision-making for all important choices.

2) CRAFT Connection

This pillar most directly embodies the CRAFT Agentic dimension, the meta-cognitive wisdom to direct AI while keeping human agency: knowing what to delegate, what to oversee, and how to evaluate outcomes. Directed writing cultivates responsibility for every element, since you cannot claim authorship if you passively accept AI's choices without evaluating them (Agentic). Deliberate choices about what to include and how to present it reflect your priorities and maintain authentic au-

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thorship even when AI-generated elements are used (Truthful, Agentic).

3) Tips for Directed Writing

Effective prompting. As Handa et al. [55] demonstrate, quality prompts enable growth mindset-supportive feedback. Be specific: provide templates, formats, and style preferences. Request specific types of feedback. The more directed your prompts, the more useful AI's output.

Figures and tables with purpose. Every visual element should serve a clear purpose. Figures should be self-explanatory with informative captions. As Edward Tufte [56] emphasizes, data graphics must address "Compared to what?", providing context and meaningful comparison. AI can generate visualizations, but you must judge whether they effectively communicate.

Sound design principles. Following Tufte's principles, ensure visual elements convey complex information efficiently. Use ordering schemes (resemblance, contiguity, cause-effect) that align with your narrative. AI can create figures; you must ensure they have explanatory power.

Organizational consistency. Maintain consistent formatting, terminology, and structural patterns. AI can help enforce consistency, but you must define what consistency means for your document.

Meta-cognitive monitoring. Regularly step back and ask: Am I directing this process, or is AI directing me? Are my choices intentional or merely convenient? This self-awareness distinguishes directed authorship from passive assembly.

The explainability test. Can you explain and defend every design choice? Why this figure here? Why this organizational structure? If your answer is "AI suggested it," you lack sufficient oversight.

4) Common Pitfalls

Passive acceptance of AI output is the fundamental pitfall: letting AI drive the process rather than directing it yourself. This produces documents you've assembled rather than authored. Other failures include superficial prioritization of speed (rapid text generation tempts writers to prioritize speed over depth, directed writing requires deliberate engagement), unclear authorship boundaries (failing to track which content is AI-generated versus human-created blurs authorship and makes evaluation impossible), and over-reliance on AI for decisions (using AI to make judgments that require human expertise abdicates authorial responsibility). AI can identify design issues, but intentional oversight requires your judgment and explicit decision-making.

5) Directed Writing Self-Assessment

The following boxes provide manual and AI-assisted **Directed Writing** audit tools:

CRAFT Assessment: Directed Design Quality

Test whether your work has directed design:

(a) Can you explain the rationale behind every major design choice?
(b) Have you actively directed AI rather or passively accepted output?
(c) Do you maintain awareness of human versus AI contributions?

CRAFTED Assessment: Directed Design Quality

Evaluate your output:

(a) Does every visual element serve a clear purpose?
(b) Is formatting consistent and professionally appropriate?
(c) Do organizational choices enhance clarity and impact?

DIY: AI-Assisted Directed Design Quality Audit

Use this prompt with your favorite GenAI tool:
"Analyze this document's design and organization. Identify: (1) visual elements (figures, tables) and whether they effectively serve their purpose, (2) organizational choices and whether they enhance or hinder communication, (3) formatting or layout consistency issues, (4) places where design choices seem arbitrary rather than intentional. Evaluate whether the document reflects clear human direction or merely AI-style assembly."

Reflect: Do I agree with these design assessments? Can I explain why I made the choices I did? If I cannot defend my design decisions, I need more intentional oversight.

I. CRAFTED Integration: The Seven Pillars Working Together

The seven CRAFTED pillars are not isolated dimensions but interconnected aspects of effective communication. Strong coherence supports flow; flow enables reader focus; reader focus requires accuracy; accuracy builds on top-down organization; top-down thinking provides structure for espirited expression; espirited writing requires directed oversight.

In practice, improving any pillar often strengthens others. When you work on coherence, you develop consciousness of logical relationships that improves flow. When you enhance reader focus, you cultivate relational awareness that makes your writing more espirited. When you improve accuracy, you develop the verification habits that support all other pillars.

Table III shows how each CRAFTED pillar connects to CRAFT dimensions, the polarity it navigates, and the capabilities it develops. This integration matrix serves as both conceptual map and practical tool, use it to understand how your work on any pillar contributes to overall development.

The path forward is clear: neither pure outside focus (risking AI slop) nor pure inside focus (risking communication failure) suffices. Excellence in AI-augmented writing requires the multiplex CRAFT(-ED) approach, simultaneously attending to formation and expression, capability and artifact, becoming and producing. The following sections address ethics, related work, and implications of this framework.

VI. Ethics and Responsible Use: Navigating Responsibility as Polarity

Ethical considerations in AI-augmented writing resist simple resolution. Rather than seeking definitive answers to "Is AI use ethical?", polarity management helps us navigate persis-

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tent ethical tensions [57]. This section reframes key ethical challenges as polarities requiring ongoing navigation, demonstrating how the CRAFT(-ED) framework offers guidance for responsible practice.

A. Ethics as Polarity Navigation

Traditional approaches to AI ethics in education often frame questions as binary: Is using AI cheating or not? Should we allow AI or ban it? These either-or framings miss the essential insight that ethical AI use involves navigating ongoing tensions between legitimate competing values. As Resnik and Hosseini [58] argue, the scientific community urgently needs new guidance for AI use in research, because existing frameworks were not designed for tools that can generate plausible but unreliable content at scale. Their call for responsible AI use principles, including researcher accountability for AI-related errors, transparent disclosure, and human oversight, aligns closely with the CRAFT(-ED) framework's emphasis on epistemic integrity and directed oversight.

Consider authorship: writers need both individual ownership (claiming their contribution) and collaborative support (acknowledging assistance). Pure individualism denies the collaborative nature of all scholarship; pure collectivism erodes individual responsibility. Neither pole alone suffices, ethical practice requires navigating between them.

The CRAFT(-ED) framework embodies this ethical navigation. The Truthful dimension of CRAFT addresses epistemic integrity and authentic representation. The Accuracy pillar of CRAFTED addresses verification and evidence standards. Together, they provide both philosophical grounding (inside) and practical guidance (outside) for ethical practice.

The integration matrix (Table III) provides a comprehensive overview of these interconnections. Throughout, we maintain the multiplex perspective illustrated in Figure 2: never choosing between inside and outside, always navigating both simultaneously through continuous cycles of development, production, reflection, and refinement.

B. Three Human Principles for Ethical AI Use

In the spirit of Asimov's Three Laws of Robotics, we offer three principles as candidate guidance rather than hard rules. First introduced in the author's IEEE *Potentials* article [59] and reformulated here through the polarity lens, they are heuristics for navigating the tension between leveraging AI and preserving human development; they are meant to provoke reflection and invite refinement, not to legislate.

Three Human Principles for AI-Augmented Writing

First Principle (Equality): When AI is readily accessible, your work should be no worse than that produced by AI alone.
Second Principle (Differentiation): When AI is present and you don't want to be made irrelevant, maintain your competitive advantage through irreplaceable human capabilities.
Third Principle (Capability Preservation): After becoming users of AI, don't let your inherent unaided capabilities decline.

1) The Equality Principle

The Equality Principle recognizes the AI Support pole's upside (accessibility, efficiency) while guarding against its downside (producing inferior work through misuse). However, it must be balanced with the Human Voice pole, you should also be developing capabilities that enable work *beyond* what AI alone could produce. The polarity: leverage AI's baseline quality while cultivating distinctive human contribution. *Navigation guidance:* Use AI to establish quality baselines, but add genuine human insight, domain expertise, and authentic voice that AI cannot provide. The goal is not AI-equivalent work but AI-enhanced work that reflects both AI's strengths and your unique contributions.

2) The Differentiation Principle

The Differentiation Principle emphasizes the Human Voice pole's upside (distinctive contribution) while avoiding its downside (rejecting useful tools). Balance requires both leveraging AI and cultivating capabilities AI cannot replicate: deep domain understanding, creative synthesis, ethical judgment, authentic relationship with readers. The polarity: use AI for productivity while developing irreplaceable human expertise. *Navigation guidance:* Identify what makes your contribution valuable beyond AI's capabilities. Develop these differentiators deliberately: domain expertise, critical judgment, creative insight, authentic voice, ethical reasoning. Use AI to handle routine tasks while focusing human effort on high-value contributions only humans can provide.

3) The Capability Preservation Principle

The Capability Preservation Principle directly addresses the automation paradox [34], [35], as AI becomes more capable, maintaining human capability becomes more critical. This principle embodies the core polarity tension: use AI for productivity while preserving and developing capability. Over-reliance produces skill atrophy; complete rejection misses productivity gains. The polarity: AI-enabled efficiency alongside deliberate capability development. *Navigation guidance:* Establish regular AI-free practice. Maintain skills that AI doesn't exercise. Monitor for warning signs of over-reliance (inability to work without AI, declining unaided performance, accepting outputs you don't understand). Use AI as scaffold, not replacement.

C. Key Ethical Challenges as Polarities

We now examine major ethical challenges through the polarity lens, showing how each represents ongoing tension rather than problems to solve.

1) Authorship and Attribution

a) The Polarity: Individual ownership ↔ collaborative acknowledgment. Scholarship has always involved building on others' work, but AI collaboration challenges traditional authorship boundaries. Pure individualism denies collaborative reality; pure collectivism erodes individual responsibility.

b) Upside of Individual Ownership: Clear accountability, responsibility for claims, authentic contribution, development

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TABLE III: CRAFT–CRAFTED Integration Matrix: A Comprehensive Navigation Guide

CRAFTED Pillar	Polarity Navigation	Primary CRAFT Dimension(s)	Key Capability Developed	Self-Assessment Focus
1. Coherence	AI-suggested structure ↔ human understanding	Consciousness	Epistemic awareness of logical relationships and conceptual integration	Can you explain why each section connects as it does?
2. Reader Focus	AI polish ↔ human empathy	Relational	Genuine engagement with readers and scholarly communities	Do you genuinely understand your audience's needs?
3. Accuracy	AI efficiency ↔ human verification	Consciousness + Truthful	Critical evaluation skills and epistemic integrity	Have you personally verified every claim?
4. Flow	AI connection ↔ human integration	Consciousness + Agentic	Understanding transitions and making deliberate organizational choices	Do transitions reflect actual logical relationships you understand?
5. Top-Down	AI details ↔ human framing	Agentic + Formative	Strategic ownership and developing judgment about purpose	Can you state your core message and defend your structure?
6. Espirited	AI formulaic patterns ↔ human creativity	Agentic + Formative	Self-authorship and developing distinctive voice	Does this writing reflect your authentic perspective?
7. Directed	AI automation ↔ human oversight	Agentic	Meta-cognitive judgment and intentional agency over process	Are you directing AI, or is AI directing you?

of genuine capability. These align with CRAFT dimensions of Agency, Responsibility, and Truthfulness.

c) Downside if Overemphasized: Denies legitimate collaborative nature of scholarship, creates false impression of solitary genius, may lead to rejecting useful tools that could enhance work.

d) Upside of Collaborative Acknowledgment: Recognizes that all scholarship builds on others, enables appropriate use of AI tools, honest about knowledge creation process. Aligns with CRAFT dimension of Relational.

e) Downside if Overemphasized: Diffuses responsibility to point where no one is accountable, enables claiming credit for work one doesn't understand, obscures individual contribution.

f) Navigation Guidance: Maintain clear boundaries between your contribution and AI's. Be transparent about AI's role while taking full responsibility for the final work. Follow disciplinary norms for attribution while ensuring you can explain, defend, and stand behind every element. The Truthful dimension of CRAFT provides guidance: never misrepresent your understanding or contribution.

2) Verification and Trust

a) The Polarity: AI efficiency ↔ human verification. AI enables rapid information synthesis, but cannot guarantee accuracy. Pure trust produces hallucinations and errors; complete skepticism makes AI unusable.

b) Upside of AI Efficiency: Rapid literature search, pattern recognition across large corpora, identification of relevant sources, initial synthesis. These support the CRAFTED pillar of Accuracy in its research phase.

c) Downside if Overemphasized: Hallucinated citations, plausible-sounding falsehoods, misrepresented arguments, false confidence from polished presentation. This is the path to AI slop.

d) Upside of Human Verification: Domain expertise to judge plausibility, understanding of source quality, disciplinary knowledge, accountability for claims. Develops CRAFT dimensions of Consciousness and Responsibility.

e) Downside if Overemphasized: Time-intensive checking, potentially missing relevant sources AI could find, confirmation bias in verification.

f) Navigation Guidance:: Use AI for efficiency but verify everything. Never trust AI as authoritative source. The Accuracy pillar provides specific guidance: verify every citation, check every fact, ensure every source exists and makes the claims attributed to it. As Mittelstadt et al. [52] recommend, use AI as zero-shot translator, not knowledge source.

3) Accessibility and Equity

a) The Polarity: Equal access to tools ↔ unequal baseline capabilities. AI democratizes access to writing assistance, but may widen gaps between those with strong foundational skills and those without.

b) Upside of Equal Access: Helps multilingual learners, supports students with disabilities, reduces barriers for those lacking privileged educational backgrounds. AI can level playing fields in important ways.

c) Downside if Overemphasized: May mask rather than address underlying capability gaps, creates dependency rather than developing independence, potentially widens capability gaps while narrowing output quality gaps.

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d) Upside of Foundational Capability Development: Ensures students develop genuine understanding and skills, prevents dependency, builds long-term capacity for independent work.

e) Downside if Overemphasized: Maintains existing inequities, privileges those with strong preparation, ignores legitimate accessibility needs, rejects tools that could genuinely help.

f) Navigation Guidance:: This polarity requires different emphasis for different populations. For those with strong skills, emphasize capability preservation (the Capability Preservation Principle). For those facing genuine barriers, emphasize AI as accessibility tool while simultaneously building foundational competence. The key: use AI to reduce barriers while developing, not replacing, human capability. The Formative dimension of CRAFT reminds us that the goal is becoming capable, not merely producing outputs.

4) Transparency and Privacy

a) The Polarity: Full disclosure ↔ privacy protection. Ethical practice requires transparency about AI use, but complete disclosure may reveal proprietary methods or personal information.

b) Upside of Full Disclosure: Enables assessment of work's authenticity, builds trust with readers, models honest practice, supports scholarly integrity. Aligns with CRAFT dimension of Truthfulness.

c) Downside if Overemphasized: May reveal competitive advantages, could expose proprietary methods, potentially invites unfair judgments based on tool use rather than work quality.

d) Upside of Privacy Protection: Protects intellectual property, prevents discrimination, maintains competitive position, respects personal choices about methods.

e) Downside if Overemphasized: Enables misrepresentation, obscures authentic contribution, makes assessment of genuine capability impossible.

f) Navigation Guidance: Follow disciplinary and institutional norms for disclosure. Be transparent about AI's general role without necessarily detailing every prompt. The test: could you defend your work's authenticity if questioned? The Directed pillar emphasizes maintaining clear awareness of human versus AI contributions, transparency begins with your own clear understanding.

5) Skill Development Versus Productivity

a) The Polarity: Efficiency through AI ↔ capability through struggle. Learning often requires productive struggle, but AI enables bypassing difficulty. Pure efficiency may prevent learning; pure struggle ignores legitimate productivity gains.

b) Upside of AI Efficiency: Faster drafting, help with routine tasks, ability to focus on higher-order challenges, accessibility for those with specific barriers. Supports productivity without sacrificing quality.

c) Downside if Overemphasized: Bypasses developmental struggle, prevents skill formation, creates dependency, leads to cognitive debt [16]. This produces the outside-without-inside failure mode.

d) Upside of Capability Development: Builds lasting skills, develops genuine understanding, creates independence, enables future sophisticated AI use. Embodies all CRAFT dimensions.

e) Downside if Overemphasized: Unnecessarily slow and difficult, maintains inequitable barriers, rejects legitimate assistance, romanticizes struggle for its own sake.

f) Navigation Guidance: Context matters. For foundational skills, emphasize struggle and capability development. For advanced applications, leverage AI for efficiency while maintaining oversight and understanding. The Formative dimension of CRAFT reminds us: the goal is becoming capable practitioners, not merely producing current outputs. Use AI to handle routine aspects while focusing human effort on capability-building challenges.

D. Institutional and Policy Implications

Institutions navigating ethical AI use should adopt polarity-informed policies that recognize ongoing tensions rather than seeking permanent solutions, articulating a Greater Purpose (developing capable writers who produce excellent work) and a Deepest Fear (AI-dependent students who cannot work independently) rather than issuing blanket prohibitions or permissions. We develop the institutional implications in detail, including policy design, assessment reform, honor-code evolution, and supporting infrastructure, in Section VIII; the ethical point here is simply that policy is itself best framed as navigation support rather than as a fixed rule.

E. Early Warning Signs and Leveraging Actions

Following Johnson's SMALL framework, we identify early warning signs of unhealthy pole emphasis and leveraging actions to restore balance.

1) Early Warning Signs of Over-Reliance on AI

Warning signs that writers may be over-emphasizing the AI Support pole include inability to work productively without AI access, accepting AI outputs without understanding, declining performance on unaided tasks, difficulty explaining or defending their work, and producing work in domains they don't understand. These signs indicate unhealthy drift toward the outside-without-inside failure mode.

2) Leveraging Actions to Restore Balance

To counteract over-reliance, writers should establish regular AI-free practice periods to maintain unaided capabilities, require self-explanation of all AI-generated content before acceptance, maintain capability development through deliberate practice in foundational skills, use dual assessment tools to evaluate both inside capability and outside artifact quality, and monitor for signs of skill atrophy while adjusting AI use accordingly.

3) Early Warning Signs of Under-Utilizing AI

Warning signs that writers may be under-emphasizing the AI Support pole include unnecessary inefficiency in routine

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tasks, missing opportunities for accessibility support, rejecting AI on principle rather than judgment, output quality below what AI-enhancement could achieve, and inequitable burden on those facing language or other barriers. These signs indicate unhealthy drift toward rejecting legitimate productivity gains.

4) Leveraging Actions to Restore Balance

To counteract under-utilization, writers should experiment with AI for routine aspects while maintaining oversight of critical elements, use AI to reduce genuine barriers while simultaneously building capability, learn effective prompting and AI direction techniques, distinguish between tasks requiring formative struggle and those suitable for AI support, and regularly assess whether rejection serves learning goals or merely preserves familiar but less effective methods.

F. Ethical Practice: A Continuous Navigation

Ethical AI use in writing is not a problem to solve but a wisdom practice to cultivate continuously. The CRAFT(-ED) framework offers both philosophical grounding (through CRAFT dimensions) and practical guidance (through CRAFTED pillars) for this ongoing navigation. The test of ethical practice is not whether you used AI, but whether you can answer affirmatively to the questions in the Ethical Practice Self-Assessment box. These questions operationalize the ethical dimensions of the CRAFT(-ED) framework, transforming abstract principles into concrete self-assessment. Ethics in AI-augmented writing emerges from the same multiplex navigation that characterizes all aspects of the framework: simultaneously attending to inside formation and outside expression, capability and artifact, becoming and producing.

Ethical Practice Self-Assessment

Consciousness: Do I genuinely understand every major claim in my work?

Responsibility and Agency: Can I explain, defend, and take responsibility for all content?

Truthfulness: Have I been transparent about AI's role and my contribution?

Accuracy: Have I verified all factual content and citations?

Relational Engagement: Does my work reflect genuine engagement with ideas and sources?

Formative Development: Am I developing capabilities, not just producing outputs?

Directed Oversight: Have I directed AI intentionally rather than passively accepted outputs?

VII. Related Work: Positioning CRAFT(-ED) in the Scholarly Landscape

Having developed the framework, we now situate it within existing scholarship. We organize prior work around the inside-outside paradox: most contributions illuminate one pole or the other, whereas CRAFT(-ED) is distinctive in addressing both at once.

A. Outside-Focused Research: AI's Impact on Writing Outputs

A substantial literature examines how AI affects writing quality and productivity, the outside pole of our paradox.

Lingard [30], [60] examines ChatGPT's capabilities and limitations in academic writing, noting both its utility for language editing and the risk of over-reliance, but focusing on output quality rather than writer development. Guo et al. [31], [61] study how creative writers integrate AI while preserving their values, emphasizing co-creation and intentionality; although they gesture toward inside concerns, their primary focus remains the creative process and its outputs.

Meta-analyses by Liu et al. [18] and Deng et al. [19] show that appropriately scaffolded AI use can enhance learning outcomes, yet they measure learning largely through output quality rather than capability development. Wang et al. [32] examine student learning from ChatGPT use, surfacing both benefits and challenges but again emphasizing measurable outputs over the inside dimension of formation.

Collectively, these studies provide crucial evidence about AI's impact on writing quality while addressing primarily the outside pole. CRAFT(-ED) extends this work by maintaining output quality while simultaneously ensuring capability development.

B. Inside-Focused Research: Capability and Formation

A complementary literature examines how AI affects writer development and capability, the inside pole of our paradox. Kosmyna et al. [16] document the accumulation of "cognitive debt" when students over-rely on AI assistants, showing how apparent productivity gains can mask capability erosion, while Zhai et al. [17] review evidence that excessive AI use can lead to skill atrophy. The automation literature sharpens the point: Bainbridge's [34] ironies of automation show how greater automation makes the human contribution more critical even as opportunities to maintain skill diminish, a dynamic that Simkute et al. [36] document in GenAI contexts and that Zohar, Bloom, and Inzlicht [14] extend by arguing that the friction AI removes is often developmentally necessary for learning, meaning-making, and social connection.

These concerns rest on a deeper philosophical foundation, which we draw on directly rather than treating it as a separate tradition. The capability approach distinguishes what people possess from what they are able to do and be: Sen [22], elaborated by Nussbaum [23], reframes the goal as writers becoming capable practitioners rather than merely holding quality documents, and Biesta [24], [44] distinguishes qualification, socialization, and subjectification, of which the last is most threatened by automation. Sennett's [40] account of craftsmanship adds that the making process matters beyond the outputs it yields. CRAFT(-ED) operationalizes these insights through its five inside-focused dimensions.

1) The Limits of Automation and the Need for Human Judgment

A related line of work cautions against the uncritical, wholesale use of automation by exposing its limits. Birhane's [62] multi-method analysis of AI's pitfalls, including dataset audits that reveal embedded structural injustices, shows that the ambiguities AI systems struggle with are precisely the complexities that demand sustained human engagement and judgment. Crawford's philosophical work [4], [37] on attention, agency, and skilled engagement clarifies what is at stake in the inside dimension and how technology shapes human capacity,

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even though it was not developed specifically for writing or AI. Together, these works underscore that automating away the difficult parts of thinking is neither neutral nor always desirable.

2) AI and Illusions of Understanding

A growing body of work addresses the epistemic risks of AI in knowledge production. Messeri and Crockett [12] identify how AI tools create "illusions of understanding" in science, a culture that produces more while comprehending less, and their subsequent analysis [13] warns that large language models make plausible-sounding text trivially easy to produce. Topaz et al. [50] supply striking evidence: an audit of 2.5 million biomedical papers finds fabricated citations embedded in the peer-reviewed literature at scale, with the rate accelerating since 2024. Resnik and Hosseini [58] argue that the research community urgently needs new ethical guidance for AI use, and Bakhshi [49] formalizes a related concern as *epistemic-rhetorical miscalibration*, in which models express more confidence than their knowledge warrants. These contributions reinforce a central premise of CRAFT(-ED): producing polished output without genuine understanding is not merely an educational concern but a systemic risk to the integrity of scholarship, directly instantiating our "outside without inside" failure mode.

Taken together, the inside-focused literature documents what is lost through over-reliance but offers limited guidance for navigating the tension between capability and productivity. CRAFT(-ED) supplies concrete guidance for holding the inside pole while leveraging the outside pole's benefits.

C. Polarity Management and Paradox Theory

Johnson's polarity management framework [9], [10], [43] distinguishes problems to solve from polarities to manage; though widely applied in organizational settings, its application to GenAI integration in education is novel. Smith and Lewis's paradox theory [25], [26] offers complementary insights on both-and thinking, and Schumacher's [21] distinction between convergent and divergent problems grounds why AI integration resists simple solutions. CRAFT(-ED) adapts these frameworks for AI-augmented writing, operationalizing the insight that divergent problems require ongoing navigation rather than permanent fixes.

D. Writing Pedagogy, Human-AI Interaction, and Engineering Education

Classic writing guides [27]–[29], [45] provide foundational principles developed in pre-AI contexts that CRAFT(-ED) updates for the GenAI era. Shneiderman's human-centered AI [63], [64] contributes design principles that keep human control central, and Vaccaro et al.'s [65] systematic review identifies when human-AI combinations prove useful. Within engineering education, writing has been identified as essential [66] and recent work examines AI's broader implications [67]; CRAFT(-ED) emerges from this context but extends across disciplines.

E. The CRAFT(-ED) Contribution: Multiplex Integration

The literatures surveyed above each illuminate one pole of the paradox or one adjacent tradition, but rarely both poles together. CRAFT(-ED) does not replace any of these traditions; it synthesizes and extends them through the polarity-management architecture. We make the positioning explicit by comparing the framework, in turn, against four adjacent traditions, namely AI literacy, self-regulated learning, human-AI collaboration, and AI-augmented writing, stating for each what it contributes and what our integration adds. What ties these contrasts together is the multiplex reading itself: CRAFT(-ED) does not merely juxtapose these traditions but orders them along the horizontal (both-and), vertical (formation ranked above fluency), and dynamic (feedback over time) dimensions of multiplexity, and it is this ordering, rather than any borrowed component, that constitutes the integrative contribution none of the traditions makes on its own.

AI literacy frameworks, such as Long and Magerko's competency model [68], define what users need to understand *about* AI systems, namely the critical appraisal of AI capabilities, limitations, and societal implications. CRAFT(-ED) presupposes this awareness but extends it into a domain-specific framework for high-stakes practice: it addresses not just what writers should know about AI but how they should navigate AI integration in academic writing as an ongoing, context-sensitive tension rather than a skill to acquire once.

Self-regulated learning (SRL) frameworks [39] describe the metacognitive processes by which learners set goals, monitor progress, and regulate their own cognition and behavior. The Agentic and Formative dimensions of CRAFT draw substantially on SRL insights, particularly self-monitoring and adaptive regulation. CRAFT(-ED), however, embeds these processes within the specific polarity created by AI assistance, where the structural risk is not merely poor self-regulation but the systematic *delegation* of regulation to the AI itself. The polarity management architecture clarifies why SRL interventions alone are insufficient in AI-augmented contexts: writers may believe they are self-regulating while the AI is in fact doing the regulatory work.

Human-AI collaboration research [65] examines the conditions under which human-AI combinations outperform either alone. CRAFT(-ED) aims to provide the practitioner-level framework for approaching those conditions in writing specifically: the polarity navigation cycle operationalizes the ongoing assessment and adjustment that effective collaboration requires.

AI-augmented writing research [30]–[32] examines how writers integrate generative-AI tools into drafting and revision, documenting gains in fluency and efficiency alongside risks of over-reliance, homogenized prose, and diminished ownership. This work characterizes the writing process and its products with increasing precision, but it largely brackets the developmental question of what the writer *becomes* through sustained AI use. CRAFT(-ED) supplies that axis: building on Bereiter and Scardamalia's distinction between knowledge-telling and knowledge-transforming composition [69], the insight that genuine writing restructures the author's own understanding

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rather than merely transcribing it, it pairs artifact-level assessment (CRAFTED) with formation-level assessment (CRAFT) in a single architecture grounded in polarity management logic. The distinguishing move is to treat AI integration as a persistent tension requiring ongoing navigation rather than a challenge admitting a once-and-for-all practical solution.

A fifth, contemporary body of work is complementary rather than overlapping: AI-use policy frameworks in educational assessment, such as the five-level AI Assessment Scale [70], govern what AI use is permitted on a given task, whereas CRAFT(-ED) addresses what the writer becomes through use. Permission frameworks and formation frameworks need each other; neither alone closes the gap between sanctioned use and developed capability.

VIII. Discussion: Implications for Research, Practice, and Design

The CRAFT(-ED) framework offers both theoretical integration and practical guidance for navigating AI-augmented writing's fundamental challenges. This section explores implications for research, pedagogy, institutional policy, and technology design, while acknowledging limitations and identifying future directions.

A. For Research: Advancing Understanding

The framework opens several research directions that extend current scholarship.

1) Empirical Validation Studies

While we provide conceptual framework grounded in established theory, empirical studies should test whether polarity-informed interventions improve outcomes. Key questions include whether students using CRAFT(-ED) self-assessment tools maintain better capability-artifact balance than control groups, whether explicit polarity framing reduces over-reliance while maintaining productivity, and whether we can measure both inside (capability) and outside (output) dimensions reliably. Future longitudinal studies tracking students through polarity-informed curricula can reveal whether early AI use affects long-term skill development and whether explicit navigation frameworks enable sustained capability development alongside productivity gains.

2) Assessment Instrument Development

We need validated instruments measuring both CRAFT capabilities and CRAFTED artifact quality. Traditional rubrics capture output quality; we need parallel instruments assessing understanding, agency, and authentic contribution. Research should develop and validate self-assessment tools for each CRAFT dimension, observer rubrics distinguishing genuine understanding from sophisticated assembly, portfolio assessment methods capturing developmental trajectories, and analytics identifying patterns of healthy versus unhealthy AI use. These instruments would enable systematic research on capability development in AI-augmented contexts while supporting pedagogical practice.

Operationalizing the framework's core constructs will require careful instrument design. Epistemic agency (Agentic dimension) could be assessed through think-aloud protocols during revision, coding how often writers critically evaluate rather than simply accept AI suggestions. Capability development (Formative dimension) could be tracked through longitudinal writing portfolios assessed under blinded AI-restricted conditions at intervals, revealing whether underlying ability grows or atrophies over time. Authentic authorship (Espirited pillar) could be operationalized through attribution studies examining whether readers can identify the writer's voice across AI-assisted and unassisted texts. Healthy AI reliance (across all dimensions) could be measured through behavioral logging of prompt types, editing depth post-generation, and explanation accuracy tests in which writers are asked to explain AI-generated passages without access to the original prompt. The inside-outside paradox itself could be operationalized as the correlation between independently assessed capability scores (CRAFT) and artifact quality scores (CRAFTED), with the prediction that polarity-informed interventions will improve both simultaneously rather than trading one off against the other. Educational assessment practice also offers ready-made process instruments for several of these constructs: AI-use disclosure logs, staged drafts produced under declared AI-use regimes, and oral defenses all generate observable traces of healthy or unhealthy reliance, and the sanctioned-use level of an assignment (for example, its AI Assessment Scale level [70]) provides a directly manipulable variable for intervention studies. In assessment terms, the dual-assessment architecture is a validity claim [71]: it asks evaluators to warrant judgments about the writer, not only about the text.

3) Cross-Disciplinary Adaptation

Our framework draws heavily on engineering education contexts. Research should examine how it translates to humanities, social sciences, professional writing, and other domains. Key questions include whether different disciplines require different polarity maps, whether some CRAFT dimensions or CRAFTED pillars are more critical in certain fields, and how disciplinary conventions affect navigation strategies. Future comparative studies across disciplines can reveal both universal principles and context-specific adaptations, strengthening the framework's theoretical foundations while improving practical utility.

4) Extension to Other AI-Augmented Domains

The polarity management approach and multiplex thinking underlying CRAFT(-ED) may apply beyond writing. Research could explore whether similar frameworks help navigate AI-augmented design, problem-solving, ethical reasoning, or creative work. This would test whether the inside-outside paradox represents a general pattern in AI-augmented expertise or is specific to writing contexts.

B. For Pedagogical Practice: Enabling Wise Navigation

Educators can implement CRAFT(-ED) principles through multiple pedagogical strategies.

1) Assignment Design

Structure assignments to explicitly navigate both poles rather than forcing binary choices. Instead of prohibiting AI or permitting unrestricted use, create assignments requiring demonstrable capability alongside quality outputs. Require writers

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to explain key concepts independently before using AI, ask for documentation showing how AI suggestions were evaluated and adapted, include reflection components addressing CRAFT dimensions explicitly, and design deliverables requiring human judgment AI cannot provide. Assignments become practice in navigation rather than compliance or prohibition exercises.

2) Scaffolded AI Integration

Provide graduated introduction to AI tools aligned with capability development. Early in courses or programs, emphasize CRAFT dimensions with limited AI use, building foundational capabilities. As capabilities develop, gradually increase AI integration while maintaining explicit oversight. This prevents premature automation while enabling eventual sophisticated AI use. The metaphor: learning to drive requires first understanding traffic, vehicle operation, and judgment before enabling cruise control or other automation. Similarly, writing capability must precede sophisticated AI augmentation.

3) Metacognitive Development

Use self-assessment tools (provided throughout Section V) to develop metacognitive awareness. Have students regularly assess both their capabilities and their outputs using dual assessment framework. This develops the meta-cognitive oversight essential for wise navigation, what Johnson calls the "Learning" step in SMALL framework. Metacognitive reflection transforms AI use from unreflective habit into intentional practice, developing the wisdom required for ongoing navigation.

4) Modeling and Professional Development

Faculty need support navigating their own AI use before guiding students effectively. Professional development should model polarity thinking rather than presenting binary choices, provide opportunities for faculty to explore their own AI-augmented practice, create communities of practice for sharing navigation strategies, and develop shared polarity maps for disciplinary contexts. When faculty understand AI integration as polarity requiring navigation, they can guide students more effectively than when operating from prohibition or uncritical adoption.

C. For Institutional Policy: Beyond Binary Choices

Institutions navigating AI policies should adopt polarity-informed approaches recognizing ongoing tensions rather than seeking permanent solutions.

1) Policy as Navigation Support

Rather than blanket prohibitions or permissions, provide frameworks for navigation. Effective policies should articulate the Greater Purpose (developing capable writers producing excellent work), name the Deepest Fear (AI-dependent students unable to work independently), provide action steps for maintaining a healthy balance, offer assessment frameworks addressing both capability and output, and enable periodic policy revision as AI capabilities and understanding evolve. Policies become living frameworks for ongoing navigation rather than fixed rules requiring constant revision.

2) Honor Code Evolution

Traditional honor codes often treat AI as cheating. Evolve codes to address real ethical issues: misrepresentation, lack of verification, absence of genuine understanding. Focus on epistemic integrity and authentic contribution rather than tool use per se. Honor codes grounded in CRAFT dimensions (Consciousness, Agentic, Truthful) address what actually matters ethically rather than policing technology use.

3) Assessment Reform

Traditional assessments often force binary choices between AI use and prohibition. Develop assessment approaches that require demonstrable capability alongside quality outputs, value authentic contribution and genuine understanding, enable appropriate AI use while preventing pure assembly, and assess both individual work and reflective capacity. The dual assessment framework (inside capability and outside artifact) provides practical model for such reform.

4) Support Infrastructure

Institutions should provide writing centers trained in AI-augmented writing navigation, technology infrastructure supporting appropriate AI access, professional development for faculty, shared resources like polarity maps for common writing contexts, and assessment support addressing both capability and output dimensions. Infrastructure enables wise navigation rather than merely regulating tool access.

D. For Technology Design: Supporting Capability-Aware Use

Current AI writing tools optimize primarily for output quality. Future tools might incorporate capability-development awareness.

1) Capability-Aware Interfaces

Tools could incorporate features supporting capability development: graduated assistance levels adjustable based on user capability, alerts when usage patterns suggest over-reliance or shallow engagement, prompts for self-assessment of understanding before accepting AI suggestions, features encouraging revision and adaptation rather than mere acceptance, and analytics helping users track their capability development over time. Such tools would support rather than undermine the inside dimension of development.

2) Transparency and Tracking

Tools should clearly distinguish human versus AI contributions through version tracking showing human revisions of AI-generated content, documentation of prompts and AI responses, clear marking of AI-generated versus human-created content, and export features enabling transparent documentation. Transparency features support the Directed pillar and ethical practice.

3) Deliberate Practice Scaffolds

Tools could incorporate graduated challenge levels preventing premature automation while supporting appropriate scaffolding. AI-free modes enable foundational practice, increasing assistance as demonstrated capability warrants. Challenges requiring human judgment AI cannot provide and practice exercises developing capabilities AI doesn't exercise would

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support the Formative dimension, ensuring tools develop rather than replace capability.

E. Limitations and Scope

Our framework has notable limitations requiring acknowledgment.

1) Primarily Conceptual Rather Than Empirical

As discussed in Section II, this is a Perspective paper that presents a conceptual framework grounded in evidence-based synthesis. We provide theoretical integration and practical guidance but not systematic empirical validation. The framework rests on established theories (polarity management, capability development, writing pedagogy) and is informed by recent empirical evidence (meta-analyses, cognitive debt studies, automation research), but its specific application to AI-augmented writing requires focused empirical testing. Framework articulation is a necessary precursor to meaningful empirical validation [33]; the present paper provides that articulation. Implementation studies and longitudinal research are needed to validate effectiveness, and we outline a concrete research agenda below.

2) Context-Specific Development

Development in engineering education contexts may limit transferability. Different disciplines may require adapted polarity maps. Cultural contexts beyond Western academic traditions may need different framings. The framework offers general architecture but requires contextual adaptation.

3) Rapidly Evolving Technology

AI capabilities advance quickly. Today's polarity maps may need revision as new affordances and risks emerge. The framework's strength, focusing on persistent tensions rather than specific tools, also creates challenges for concrete guidance as technology evolves. Ongoing revision will be necessary.

4) Absence of Expert Validation and Pilot Testing

The CRAFT(-ED) framework has not been subjected to expert panel review, Delphi consultation, structured peer critique by writing pedagogy specialists, or educational pilot testing in classroom settings. The framework's constituent theoretical pillars are, however, individually well-supported by empirical evidence in their respective home domains: polarity management theory is grounded in decades of organizational research [9], [10]; self-regulated learning principles have strong empirical backing across educational contexts [39]; writing pedagogy principles have been validated through extensive practice and research [27], [29]; and capability development philosophy has been extensively applied in human development research [22], [23]. What has not yet been empirically tested is their specific *integration* and application to AI-augmented writing contexts. Claims throughout this paper regarding the framework's potential to improve writing quality, support capability development, or enable balanced AI integration in these specific contexts should therefore be understood as theoretically grounded propositions requiring domain-specific empirical validation, well-motivated by the evidence base in adjacent fields, but not yet directly demonstrated for AI-augmented academic writing. This staged approach, from theoretical articulation to targeted empirical testing, is a characteristic feature of the Perspective paper genre [33], and it motivates the concrete research agenda outlined in Section VIII.

5) Individual Focus

The framework addresses individual writer navigation but gives less attention to collaborative writing, institutional dynamics, or systemic inequities. While we address equity concerns in the ethics section, deeper treatment of structural issues is needed. Future work should examine how CRAFT(-ED) principles apply to collaborative contexts and institutional transformation.

F. Future Directions: Continuing the Navigation

Priority directions for future work include *design-based research* implementing polarity-informed assignments across multiple courses and institutions, studying both immediate outcomes and long-term capability development; *shared polarity mapping* across institutions to identify common patterns and disciplinary variations, creating community resources for educators; *extension to other domains* beyond writing, applying multiplex frameworks to AI-augmented design, problem-solving, and creative work; *collaborative writing frameworks* extending CRAFT(-ED) to team contexts where navigation becomes collective rather than individual; *digital tool development* for collaborative polarity mapping, self-assessment, and capability tracking; *cross-cultural adaptation* examining how framework principles apply in diverse educational and cultural contexts; and *longitudinal studies* tracking writers through sustained polarity-informed practice to understand long-term effects on capability and productivity. A further priority is *comparative and alternative-perspective analysis*: future work should place CRAFT(-ED) explicitly alongside competing accounts of AI-assisted writing, including the growing body of evidence that documents clear learning benefits under well-designed pedagogical conditions, and test where the framework's predictions agree with, extend, or diverge from them. Because the present Perspective advances a single, evidence-based position, we judge that such a systematic comparison is better pursued as dedicated future work than folded into the argument here, where it would dilute the focus; the polarity treatment throughout already engages both poles of each contested question rather than only the cautionary one.

G. The Enduring Question

The central question of AI-augmented writing is not, "*Should we use AI?*" but "*How do we maintain dynamic equilibrium between capability development and productivity, between becoming and producing?*" This is not a question to answer once but a wisdom practice to cultivate continuously: the value of CRAFT(-ED) lies not in eliminating the tension but in making it visible, discussable, and navigable. As AI capabilities evolve and educational contexts shift, its both-and logic provides a stable foundation for adaptation.

IX. Conclusion

The CRAFT(-ED) framework addresses AI-augmented writing's fundamental challenge: how to simultaneously leverage AI's transformative capabilities while cultivating authentic human understanding, voice, and agency. As the preceding

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sections have shown, this challenge is a persistent polarity, not a problem admitting permanent solution, and excellence requires maintaining both dimensions in dynamic relationship rather than choosing between them. The framework offers concrete tools for this navigation, integrating five inside-focused CRAFT dimensions with seven outside-focused CRAFTED pillars and dual assessment instruments that evaluate capability and artifact quality together. The stakes are profound: as text generation becomes ubiquitous, what becomes precious is the ability to understand deeply, judge wisely, contribute authentically, and express effectively. These qualities cannot be automated or outsourced; they must be developed through engagement, struggle, and persistent practice. This is not a call for rejecting AI or a romantic return to pre-digital craft, but rather recognition that high performance with deep personhood requires both excellent outputs and genuine capability development. Our hope is that writers, educators, and institutions will use this framework not as prescription but as invitation to engage thoughtfully and wisely with one of the defining educational challenges of our time. When anyone can generate text, learning the CRAFT of producing genuinely CRAFTED work becomes essential, not as nostalgia for simpler times but as commitment to what makes scholarship meaningful: genuine understanding, honest representation, authentic voice, thoughtful judgment, and contribution that adds real value to human knowledge. Even in the GenAI era, authentic CRAFT matters.

Declaration of AI Use

In preparing this manuscript, the author used AI tools (ChatGPT, Claude, and Grammarly) for writing support, editing assistance, and formatting. All AI-assisted content was reviewed, verified, and revised by the author. The author takes full responsibility for the accuracy, integrity, and originality of all work presented here.

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